

Software Requirements Specifications

Hansraj Query Bridge

The Intelligent Query Resolution System

< Version 1.0 >

Submitted in partial fulfilment of the requirements of

DSC 15: Software Engineering

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of **HANSRAJ QUERY BRIDGE : Intelligent Query Resolution System**, an AI-assisted institutional query management platform designed for Hansraj College.

This document serves as a comprehensive reference for the requirements, design, development, deployment, and evaluation of the system. It establishes the expected system behaviour, user interactions, data requirements, functional requirements, AI/RAG requirements, external interfaces, and performance constraints to ensure that the developed system satisfies the intended objectives and user needs.

1.2 Scope of the Project

HANSRAJ QUERY BRIDGE: Intelligent Query Resolution System aims to provide students with a centralized platform through which they can submit queries related to administrative procedures, departmental processes, certificates, NOCs, documentation, and other college-related services. The system uses Retrieval-Augmented Generation (RAG) and a verified institutional knowledge base to provide relevant and reliable responses. When a suitable verified answer is unavailable or the system is uncertain, the query is forwarded to the appropriate administrative authority or Teacher-in-Charge (TIC) for human resolution.

The system also provides dedicated dashboards for students, administrators, and TICs to facilitate query tracking, resolution, department-wise routing, document submission, meeting scheduling, and notifications. Verified resolutions provided by authorized personnel are incorporated into the knowledge base, enabling the system to progressively handle a wider range of recurring queries.

The scope of the system includes the following major functionalities:

1. User Authentication and Role Management
2. RAG based AI-Powered Query Resolution
3. Intelligent Query Escalation and Routing
4. Knowledge Base Management
5. Role-Specific Dashboards and Management Interfaces
6. Document Management
7. Meeting and Timetable Management
8. Notifications through Email
9. Security and Data Protection
10. Query Analytics and Monitoring

The initial version of the system will not independently make official administrative decisions, modify college policies, or provide unverified institutional information. The AI will assist with information retrieval and query understanding but will not replace administrators or TICs for decisions requiring human authority.

1.3 Intended Audience

This SRS document is intended for all stakeholders involved in the development, evaluation, deployment, and use of this project. The primary audience includes:

- **Students:** End users who submit queries, receive verified responses, upload requested documents, and schedule meetings.
- **Administrators:** Users responsible for reviewing, resolving, forwarding, and managing student queries and institutional knowledge.
- **Teachers-in-Charge (TICs):** Departmental authorities who handle assigned queries, provide resolutions, request documents, and schedule student meetings.

- **Development Team:** Developers and technical team members responsible for designing, implementing, testing, and maintaining the system.
- **College Authorities:** Stakeholders responsible for evaluating, approving, and overseeing the deployment of the system.
- **Project Evaluators:** Faculty members, mentors, and idea-thon judges who assess the system's functionality, feasibility, and technical implementation.

1.4 Definitions, Acronyms and Abbreviations

Term/ Abbreviations	Definition
AI	Artificial Intelligence; technology used to understand and process user queries.
RAG	Retrieval-Augmented Generation; an approach that retrieves verified information from the knowledge base before generating a response.
TIC	Teacher-in-Charge; faculty member responsible for handling queries related to a particular department.
LLM	Large Language Model; AI model used to understand and generate natural-language responses.
RBAC	Role-Based Access Control; restricting system access according to a user's assigned role.
MVP	Minimum Viable Product; the initial version containing the essential system functionality.
SRS	Software Requirements Specification; document defining the requirements and expected behaviour of the software.
API	Application Programming Interface; a set of rules and protocols enabling different software applications to communicate and exchange data.
Knowledge Base	A collection of verified institutional questions, answers, procedures, and related information.
Semantic Search	Searching based on the meaning and context of a query rather than exact keywords.
Embedding	Numerical representation of text used to identify semantic similarity between queries.
Query	A question or request submitted by a student through the system.
Admin	Authorized administrative user responsible for managing and resolving or forwarding student queries.
Stakeholder	Any person with an interest in the project who is not a developer.
User	Reviewer or Author.

1.5 References

The following resources and technologies are referenced for the design and development of HANS Connect:

- **IEEE Software Requirements Specification guidelines** for structuring and documenting software requirements.
830-1984 - IEEE Guide for Software Requirements Specifications

- **Figma Documentation** for initial UI designing and UX development.
<https://www.figma.com/resource-library/>
- **Fast API Documentation** for backend API development.
- **Next.js, Tailwind CSS Documentation** for frontend development.
- **PostgreSQL Documentation** for relational database management.
- **pgvector Documentation** for vector storage and semantic similarity search.
- **LLM and Embedding Model Documentation** for AI-based query understanding and RAG implementation.
- **Hansraj College official website and available institutional information** as potential sources for verified college-related information.

1.6 Document Overview

This SRS document defines the functional, non-functional, technical, and performance requirements of **HANSRAJ QUERY BRIDGE : The Intelligent Query Resolution System**. It describes the system's purpose, scope, users, interfaces, system features, performance expectations, data requirements, quality attributes, and specialized AI/RAG requirements.

Readers are recommended to review the document in the following order:

- **Section 1 – Introduction:** Presents the purpose and scope of the project, intended audience, terminology, references, and an overview of the document.
- **Section 2 – Overall Description:** Provides a high-level description of the system, including its product perspective, interfaces, major functions, user classes, operating environment, constraints, assumptions, dependencies, and allocation of current and future requirements.
- **Section 3 – Specific Requirements:** Defines the detailed requirements of HANSRAJ QUERY BRIDGE. It covers external interfaces, system features, performance requirements, logical database requirements, design constraints, software system attributes, and other specialized requirements, including AI and RAG capabilities.
- **Section 4 – Appendix:** Provides supporting technical and analytical material, including detailed use cases, system architecture diagrams, AI/RAG flow, DFD's and ER diagrams.

Readers primarily interested in the system's overview should focus on Sections 1 and 2, while technical and development teams should also thoroughly review Section 3 and Appendix.

2. Overall Description

2.1 Product Perspective

HANSRAJ QUERY BRIDGE : Intelligent Query Resolution System is a centralized, web-based institutional support platform designed to improve the way students interact with the administrative and departmental authorities of Hansraj College.

The system acts as an intelligent layer between students, the administration, and Teachers-in-Charge (TICs). It combines conventional query management with an AI-powered RAG architecture, enabling the system to retrieve verified institutional information and provide relevant responses to students.

The system operates as follows:

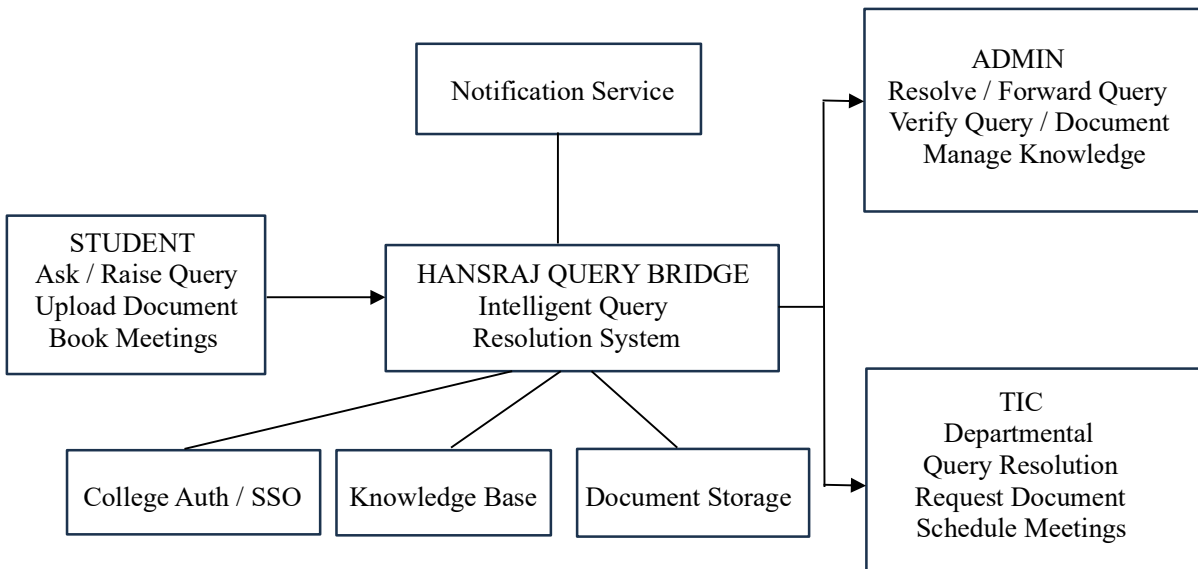


Figure 2.1: HANSRAJ QUERY BRIDGE : Product Context Diagram

HANSRAJ QUERY BRIDGE is intended to complement, rather than replace, existing college administrative processes.

The system consists of **client-side portals, gateway and API services, core query and AI/RAG services, and persistent storage components.**

The system also maintains a **continuously updated institutional knowledge base.** Verified solutions provided by authorized administrators and TICs can be incorporated into the knowledge base, allowing the system to handle recurring queries more efficiently while ensuring that uncertain or new queries remain under human supervision.

It also provides a **notification service** to inform clients about query resolutions status, document requests and meeting schedules.

2.1.1 System Interfaces

HANSRAJ QUERY BRIDGE shall interact with several internal and external system components to provide various functionalities like providing authentications, AI-powered query resolutions, data persistence, document handling, notifications, and meeting management. These interfaces enable communication between the system and the intended services required for its operation.

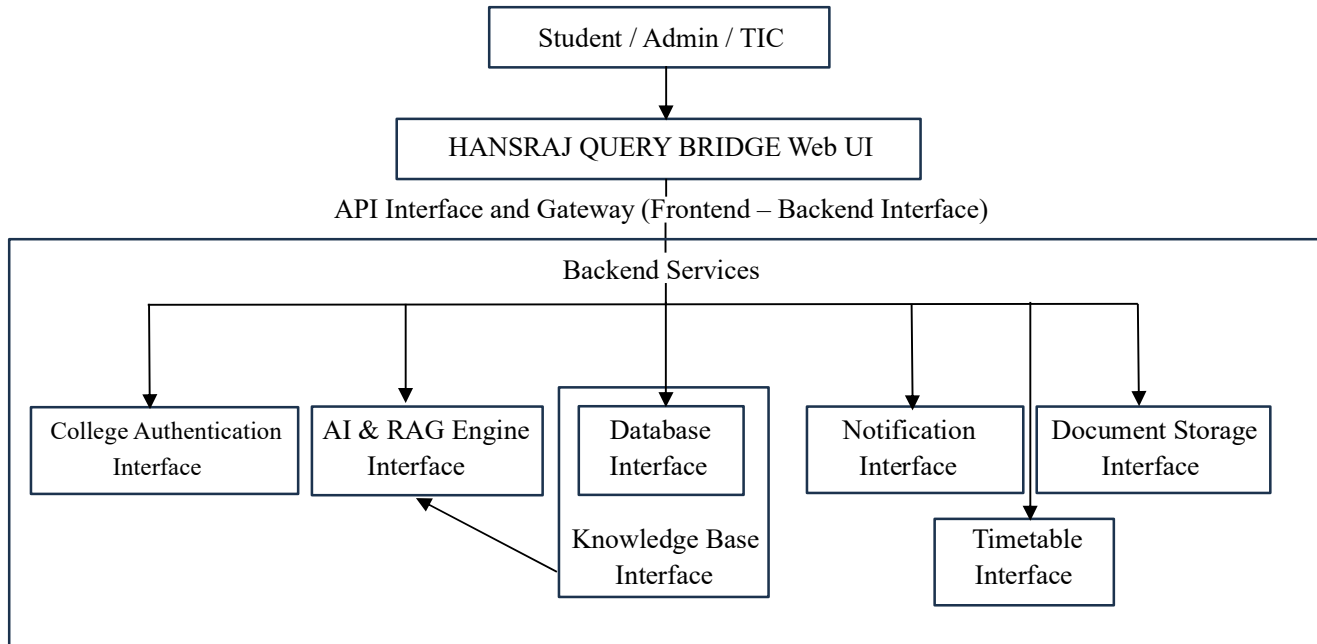


Figure 2.2: HANSRAJ QUERY BRIDGE : System Interfaces Diagram

The primary system interfaces are as follows :

A. Frontend–Backend Interface

- The web-based frontend (UI) shall communicate with the backend services through defined APIs.
- The backend shall process requests, apply authentication and authorization rules, invoke required services, and return appropriate responses to the frontend.

B. College Authentication Interface

- The system shall interface with the college's authentication mechanism, where available, to authenticate Students, Admins, and TICs.
- The authentication interface shall provide the user's identity and associated role information required for authorization and access control.

C. AI & RAG Engine Interface

- The main system shall interface with the AI/RAG subsystem to process natural-language queries, generate embeddings, retrieve relevant verified Knowledge Base information, evaluate query relevance, and generate verified and grounded responses.
- Queries for which reliable information cannot be retrieved shall be returned to the application for routing to the appropriate human authority.

D. Knowledge Base Interface

- The system shall interface with the institutional Knowledge Base to store and retrieve verified information.
- The interface shall support adding, updating, versioning, deactivating, and archiving Knowledge Base entries.
- Only active and valid information shall normally be made available to the AI/RAG retrieval process.

E. Database Interface

- The system shall interface with the PostgreSQL database for persistent storage and retrieval of system data, including users, queries, resolutions, Knowledge Base records, meetings, notifications, and audit information.
- The database shall also support vector storage through pgvector for semantic retrieval.

F. Document Storage Interface

- The system shall interface with secure file storage for storing documents submitted by students.
- The interface shall support secure upload, retrieval, access control, and lifecycle management of documents associated with authorized queries.

G. Timetable Interface

- The system shall interface with timetable data to determine relevant schedules and available time slots.
- This information shall support TICs in offering and scheduling meetings while helping prevent conflicts.

H. Notification Interface

- The system shall interface with email notification service to send notifications related to query submission, status changes, resolutions, document requests, document review outcomes, and meeting activities.

2.1.2 User Interfaces

HANSRAJ QUERY BRIDGE shall provide a role-based, web-based user interface through which Students, Admins, and TICs interact with the system.

A. Student Interface

The primary interface components for the students shall include:

1. **Login Interface:** Allows students to authenticate using the college authentication system.
2. **Student Dashboard:** Displays recent queries, current query statuses, notifications, pending document submissions, and upcoming meetings.
3. **AI Query and AI Response Interface:** Provides a chatbot-style interface where students can enter natural-language questions and may be provided with appropriate responses based on verified institutional knowledge base.
4. **Raise Query Interface:** Allows students to submit a query to the Admin when the AI cannot provide a sufficiently reliable answer or when the student requests human assistance.
5. **Query Tracking Interface:** Displays the current status, relevant timestamps, assigned authority, and resolution of submitted queries.
6. **Document Submission Interface:** Allows students to upload documents requested by a TIC and view the status of submitted documents.
7. **Meeting Interface:** Allows students to view available meeting slots, accept proposed meetings, and view scheduled or cancelled meetings.
8. **Notification Interface:** Displays important system notifications related to queries, documents, resolutions, and meetings.

B. Admin Interface

The primary interface components for the Admin shall include:

1. **Login Interface:** Allows admin to authenticate using the college authentication system.
2. **Admin Dashboard:** Displays counts and lists of new, unresolved, forwarded, and resolved queries.
3. **Query Management Interface:** Allows Admins to inspect query details, student information relevant to resolution, query history, and routing information.
4. **Query Resolution Interface:** Allows an Admin to provide a verified resolution and mark the query as resolved.
5. **Query Forwarding Interface:** Allows an Admin to forward department-specific queries to the appropriate TIC.
6. **Knowledge Base Management Interface:** Allows authorized Admins to add, update, deactivate, archive, and manage versions of verified institutional information.

7. **Notification/Status Interface:** Allows Admins to view the current state of queries and relevant system actions.

C. TIC Interface

The primary interface components for the TIC shall include:

1. **Login Interface:** Allows admin to authenticate using the college authentication system.
2. **TIC Dashboard:** Displays queries assigned to the TIC along with their current status, along with relevant student and query information.
3. **Resolution Interface:** Allows the TIC to provide a verified departmental resolution.
4. **Document Request Interface:** Allows the TIC to mark that documents are required and specify the documents and instructions for submission.
5. **Document Review Interface:** Allows the TIC to review submitted documents and mark them as accepted, rejected, or requiring correction.
6. **Timetable Interface:** Allows the TIC to view relevant timetable information and available periods.
7. **Meeting Management Interface:** Allows the TIC to offer, schedule, reschedule, or cancel meetings with students.
8. **Notification Interface:** Displays relevant query, document, and meeting notifications.

D. Interface Optimization Requirements

1. **Role-Based Interface:** Users shall see only the features and information relevant to their authorized role.
2. **Consistent Navigation:** Menus, buttons, forms, status indicators, and navigation patterns shall remain consistent across different interfaces. Users should be able to move from a query to its resolution, required documents, or meeting information without unnecessary navigation.
3. **Form Validation:** Input fields shall provide appropriate validation and meaningful error messages.
4. **Readable information presentation:** Query details, responses, instructions, and statuses should be presented in a structured and easily scannable manner.
5. **Accessibility:** Interface elements shall be designed to support users with different levels of technical familiarity and accessibility needs.

2.1.3 Hardware Interfaces

HANSRAJ QUERY BRIDGE is a web-based software system and DOES NOT require any specialized external hardware for its core functionality. The system shall operate using standard computing devices capable of running a modern web browser and accessing the internet.

A. User Device Interface

The system shall interface with standard user devices such as:

- Desktop computers
- Laptops
- Tablets
- Smartphones

The interaction between the device hardware and the system shall occur through a modern web browser. The system shall receive user input through standard hardware components such as the keyboard, mouse, touchscreen, and device display.

B. Server and Storage Hardware Interface

The application backend shall operate on a cloud computing infrastructure. Student-uploaded documents may

be stored using secure server or cloud object storage.

C. Network Hardware Interface

The system shall not require any dedicated networking hardware at the client side.

The system shall use the network interface available on the user's device and server infrastructure to communicate over the internet or institutional network.

D. Specialized Hardware Requirements

No specialized external hardware, or IoT devices shall be required for the core operation of HANSRAJ QUERY BRIDGE.

If future institutional integrations introduce specialized hardware, such hardware shall be treated as an optional extension and SHALL NOT be considered a dependency of the core system.

2.1.4 Software Interfaces

HANSRAJ QUERY BRIDGE shall interface with several software products and application services required for the system required for authentication, AI-powered query resolution, data management, document storage, notifications, and meeting management.

The following software products and interfaces are considered for the proposed system:

Software Product	Mnemonic	Version	Source	Specification Number	Purpose
Next.JS	Next.JS	14.2+	Vercel Inc.	ECMA-262 / RFC 7540	Frontend web application and user interfaces
React	React	18.3+	Meta Open Source	W3C DOM Level 4	Component-based frontend development
Python	Python	3.11+	Python Software Foundation	PEP 3333 / ASGI 3.0	Backend application development
FastAPI	FastAPI	0.111+	Sebastián Ramírez (tiangolo)	OpenAPI v3.1.0	REST API and backend service layer
PostgreSQL	PostgreSQL	16.2+	PostgreSQL Global Dev Group	ISO/IEC 9075:2023	Relational database management
pgvector	pgvector	0.7.0+	Open Source	HNSW / IVFFlat Extension Spec	Storage and similarity search of vector embeddings
Object Storage Service	AWS S3 Bucket	API v2 (2006-03-01)	AWS	RFC 2616 / S3 REST API	Secure storage of uploaded student documents
LLM / Embedding Service	GEMINI_API	1.5 Pro / Flash	Google DeepMind	JSON Schema 2020-12	Query understanding, embeddings, retrieval support and grounded response generation
Email Service	RESEND_API	v1 REST API	Resend Inc. / Twilio	RFC 5321 / RFC 5322	Sending system notifications and workflow updates

A. Interface Flexibility and Fallback Handling

- The system interfaces shall allow the system to communicate with external or supporting software components while keeping the core application modular and maintainable.
- HANSRAJ QUERY BRIDGE shall be designed so that external services can be replaced or reconfigured without requiring major changes to the core application.
- The system shall use well-defined APIs and modular service interfaces wherever practical.
- External-service failures shall be handled appropriately, with appropriate error messages, retry options where suitable, and fallback workflows.

2.1.5 Communications Interfaces

HANSRAJ QUERY BRIDGE shall use standard network communication protocols to enable secure communication between users, application services, databases, and external services.

No proprietary communication protocol shall be required.

A. Internet Communication

- Users shall access HANSRAJ QUERY BRIDGE through an internet or within the college network or from an external network, subject to the authentication and access policies of the deployment environment.
- The system shall support communication over standard TCP/IP-based networks.

B. Browser and Frontend-Backend Communication

- All communication between the user's web browser and the system shall use HTTPS (HTTP over TLS).
- Sensitive information shall not be transmitted through unencrypted HTTP connections.
- The frontend and backend shall communicate using HTTP/HTTPS-based REST APIs.
- API requests and responses shall normally use JSON for structured data exchange.
- File uploads shall use an appropriate protocol supported by the configured AWS storage service.

C. Database and AI/RAG Communication

- The backend shall communicate with the PostgreSQL database using the database's supported network protocol and a secure database connection.
- The database shall not be directly accessible to ordinary users through the public network.
- AI/RAG, embedding and any other external services shall be communicated with their supported secure APIs or protocols.

2.1.6 Memory Constraints

HANSRAJ QUERY BRIDGE shall operate as a web-based application and shall require memory resources on both the client devices and the server-side infrastructure.

The exact hardware configuration may vary depending on the number of concurrent users, volume of stored data, and deployment environment.

A. Primary Memory Constraints

- On the **client side**, HANSRAJ QUERY BRIDGE SHALL NOT require dedicated or specialized memory.
- On the **server side**, primary memory shall be used by:
 - Frontend and backend application processes.
 - API request handling.
 - Database connections.
 - AI/RAG processing, query processing and retrieval operations.
 - Authentication and session management

- Notification and background services.
- Temporary processing of uploaded documents.

Limits : The server memory requirement shall be scalable according to the expected number of concurrent users and system workload.

B. Secondary Memory Constraints

- Secondary storage shall be required for persistent system data, including :
 - User and role information.
 - Student queries and resolutions.
 - Knowledge Base entries and their version history.
 - Meeting and timetable-related records.
 - Notification records.
 - Audit and system logs.
 - Student-uploaded documents and associated metadata.
 - Application configuration and supporting files.

Limits : The storage requirement shall increase over time as the number of queries, Knowledge Base entries, documents and audit records increase.

2.1.7 Site Adaptation Constraints

HANSRAJ QUERY BRIDGE is intended to be deployed for use within **Hansraj College**, while it can be scaled to be use in any educational institution.

A. Institution – Specific Initialization

During initial deployment, the following institution-specific information shall be configured or initialized:

- Institution name and relevant institutional information.
- Departments and academic programmes.
- Academic years, semesters, and academic periods.
- Student, Admin, and TIC accounts or their connection to the institutional authentication system.
- Department-to-TIC mappings.
- Administrative roles and permissions.
- Initial Knowledge Base containing verified institutional information.
- Institutional contact and notification details.
- Timetable information, where available.
- Document types that may be requested from students.
- Applicable document upload limits and accepted file formats.

B. Knowledge Base Initialization

The Knowledge Base shall be initialized with some verified, institution-specific information before the AI/RAG query-resolution functionality is made available for routine use.

Initial Knowledge Base data may include:

- Admission and registration procedures.
- Examination-related information.
- Certificates and document procedures.
- Admin office information.
- Department-specific procedures.

- Academic schedules and important dates.
- Student services and facilities.
- Frequently asked institutional questions and verified resolutions.

C. User and Role Initialization

The installation shall provide an initialization process for configuring institutional users and their roles.

The system shall support configuration of:

- Student Credentials and institutional identifier.
- Admin Accounts.
- TIC Accounts.
- Department associations.
- Role-based permissions.
- Active/inactive user status.

D. Deployment – Specific Adaptation

The following deployment settings may need to be adapted according to the installation environment:

- Server/cloud infrastructure configuration.
- Database connection settings.
- AWS S3 Bucket configuration.
- Domain and HTTPS/TLS configuration.
- Email service configuration.
- Authentication configuration.
- AI-service configuration.
- Backup and retention settings.
- Network and firewall policies.

E. Site – Specific Configuration

For deployment at a particular institution, site-specific adaptation shall primarily be performed through configuration, database initialization, and Knowledge Base setup.

The following settings shall be configurable for a particular installation without requiring modification of the core application:

Configuration Area	Site – Specific Information
Institution	Name and institutional details
Departments	Department names and identifiers
Users	Students, Admins, TICs users
Role Mapping	Department-to-TIC and permission mappings
Knowledge Base	Verified institutional information
Timetable	Academic timetable and available periods
Documents	Permitted document types and size limits
Notifications	Institutional email configuration
Authentication	SSO/identity-provider configuration
Query Routing	Department based routing rules

F. Initialization Sequence

Installation of the Software → Institution Configuration → Department, Roles & Mapping Configuration → User and Authentication Configuration → Initial Knowledge Base Loading → Timetable Configuration → Document Storage Configuration → Notifications and Email Credentials Configuration

→ System Validation & Testing → System Operational

2.2 Product Functions

HANSRAJ QUERY BRIDGE provides the following major functions:

1. **User Authentication, Authorization and Role Management**
Provides secure login and role-based access for Students, Admins, and Teachers-in-Charge (TICs).
2. **RAG-Based AI-Powered Query Resolution**
Understands student queries using semantic search and RAG to retrieve relevant information from the verified institutional knowledge base.
3. **Intelligent Query Escalation and Routing**
Escalates queries when no reliable answer is available and allows Admins to resolve them or forward them to the appropriate TIC.
4. **Knowledge Base Management**
Stores verified query resolutions and allows authorized users to update, verify, deactivate, or archive outdated information.
5. **Student Dashboard and Query Management**
Allows students to raise queries, view responses, track query status, receive notifications, and access their query history through their dedicated dashboards.
6. **Admin and TIC Dashboards**
Provides dedicated dashboards for managing assigned, unresolved, resolved, and forwarded queries along with relevant student information.
7. **Document Management**
Enables authorized personnel to request documents and students to securely upload and submit the requested documents for review.
8. **Timetable and Meeting Scheduling**
Allows TICs to view their timetable and available slots, schedule meetings with students, and manage appointment conflicts.
9. **Notification Management**
Sends notifications for query updates, resolutions, document requests, document submissions, and scheduled meetings.
10. **Security and Monitoring**
Implements authentication, role-based authorization, input validation, secure document handling, and protection against malicious AI inputs.

2.3 User Classes and Characteristics

HANSRAJ QUERY BRIDGE has three primary user classes. Each user class has different responsibilities and access permissions within the system.

All users are authenticated and provided access according to their assigned role through Role-Based Access Control (RBAC).

User Class	Characteristics and Responsibilities
Student	Primary end user of the system. Students can log in using college credentials, ask queries through the AI chatbot, raise new queries when no verified answer is available, track query status, view resolutions, upload requested documents, receive notifications, and schedule or respond to meeting requests from TICs.

Admin	Responsible for managing institutional and administrative queries. Admins can view new and unresolved queries, resolve queries, forward department-specific queries to the appropriate TIC, manage verified knowledge - base information, monitor query progress, and review flagged or exceptional requests.
TIC	Department-level authority responsible for handling queries assigned to their department. TICs can view assigned queries and student details, provide verified resolutions, request and review documents, access their timetable, and schedule meetings with students for queries requiring offline interaction.

2.4 Operating Environment

HANSRAJ QUERY BRIDGE will operate as a web-based application accessible through standard internet-enabled devices.

The system will support the following environment:

Component	Technology / Environment
Client	Web browsers on desktops, laptops, tablets, and smartphones
Frontend	Next.js, React, and Tailwind CSS
Backend	Python with FastAPI
Database	PostgreSQL with pgvector for vector-based semantic search
AI Layer	Gemini LLM and embedding model for query understanding, semantic retrieval, and RAG-based responses
File Storage	Secure cloud (AWS S3 Bucket) storage for student documents
Communication	HTTPS and REST APIs using JSON
Notification Service	Email notification service using Resend

The system shall require a stable internet connection for communication between the client, backend services, database, AI services, and external integrations.

It shall be designed to be platform-independent and should not require specialized hardware or software on the user's device.

2.5 Implementation and Design Constraints

The development and implementation of **HANSRAJ QUERY BRIDGE** are subject to the following constraints:

- **Student Credentials Availability:** The accuracy of the system depends on the availability and correctness of official student information including college credentials and departmental mapping with TICs.
- **College Data Availability:** The system depends on the availability and correctness of official college information including administrative procedures, timetables, and departmental data.
- **AI Reliability:** The AI must not generate unverified institutional information. Responses should be based on verified knowledge retrieved through the RAG system, with low-confidence queries escalated to Admins or TICs.
- **Dynamic Information Reflection:** Changes in office timings, administrative procedures, faculty assignments, or departmental information must be reflected in the knowledge base without requiring retraining of the AI model.

- **Data Privacy and Security:** Student credentials, personal information, queries, and uploaded documents must be protected through secure authentication, role-based access control, and secure data storage. These information must only be accessible only to authorized users.
- **Human Verification:** Only responses verified/provided by authorized Admin or TIC users shall be incorporated into the institutional knowledge base.
- **External Dependencies:** Authentication, email notifications, AI services, document storage, and timetable information may depend on external services and network availability.

Further details into design constraints shall be discussed in **Section 3.5 Design Constraints**.

2.6 Assumptions and Dependencies

Assumptions

1. Students, Admins, and TICs will have valid credentials and access to the system according to their assigned roles.
2. Authorized Admins and TICs will provide accurate and verified information for resolving queries and updating the knowledge base.
3. Students will provide sufficiently clear and relevant information when submitting queries and uploading requested documents.
4. Users will have access to an internet-enabled device and a stable internet connection.
5. The system will use verified institutional information as the primary source for AI-generated responses.

Dependencies

1. The system depends on the availability of the **college authentication system** for secure user verification.
2. AI-based query processing depends on the availability of the selected **LLM and embedding services**.
3. Semantic retrieval depends on **PostgreSQL with pgvector** and the availability of the knowledge-base data.
4. Document submission depends on a secure cloud **AWS S3 services**.
5. Email notifications depend on an external **email delivery service** and **Resend/SendGrid services**.
6. Timetable and meeting scheduling may depend on the availability and accessibility of the college's **timetable data or timetable scheduling system**.
7. System performance and availability depend on the reliability of the underlying **database and external services**.

2.7 Apportioning of Requirements

HANSRAJ QUERY BRIDGE shall be developed in phases so that the core query-resolution functionality can be implemented first, while advanced or non-essential capabilities may be introduced in later versions.

A. Requirements That May Be Deferred to Future Versions

1. **Advanced Query Analytics and Reporting**
Detailed dashboards for query trends, frequently asked questions, resolution time, department workload, and AI performance.
2. **Advanced AI-Based Query Classification and Routing**
More sophisticated automatic classification of queries and intelligent routing based on department, topic, urgency, or historical resolution patterns.
3. **Multilingual Query Support**
Support for multiple languages to allow students to interact with the system in languages beyond the initial implementation.
4. **Voice-Based Query Interaction**
Speech-to-text input and voice-based responses for improved accessibility and convenience.

5. Mobile Application

Dedicated Android/iOS applications if usage patterns demonstrate a need beyond the responsive web interface.

B. Future Requirement Prioritization

Future requirements shall be prioritized based on:

- User demand and feedback.
- System usage and query volume.
- Technical feasibility.
- Institutional requirements.
- Security and privacy considerations.
- Availability of reliable external APIs or data sources.
- Performance and scalability requirements.
- Benefits relative to implementation complexity.

Advanced capabilities shall be **apportioned to future versions** based on actual institutional needs and the maturity of the system.

3. Specific Requirements

3.1 External Interface Requirements

This section specifies the detailed requirements governing the interfaces through which HANSRAJ QUERY BRIDGE exchanges information with its users and external software or hardware components. The external interfaces shall be designed to provide consistent and predictable communication between HANS Connect and its Students, Admins, TICs, and supporting external services.

It complements the high-level interface descriptions provided in **Section 2.1** by defining the expected inputs, outputs, data formats, validation rules, timing, screen organization, and interaction behaviour.

3.1.1. User Interfaces

A. Login Interface

Attribute	Specification
Name of Item	Login Interface
Description of Purpose	Provides authenticated access to HANSRAJ QUERY BRIDGE and determines the user's authorized role.
Source of Input	Student, Admin, or TIC; or configured college authentication/SSO service.
Destination of Output	Authentication service and HANSRAJ QUERY BRIDGE session-management component.
Valid Range / Accuracy	Credentials must fit to the requirements of the system authentication mechanism.
Units of Measure	Not applicable.
Timing	Authentication result should normally be displayed within a few seconds under normal network conditions.
Relationship to Other Inputs/Outputs	Successful authentication determines the dashboard and functions available to the user according to role.
Screen Format / Organization	Login page containing institutional branding, authentication fields/SSO option, Login control, and error or assistance messages.
Window Format / Organization	Single responsive login window.
Data Format	Text input and authentication tokens information handled internally.
Command Format	Login action initiated through a button interface control.
End Messages	“Login Successful” → User is redirected to the appropriate dashboard. “Login Failed” → Clear authentication error message is displayed.

B. AI Query Interface

Attribute	Specification
Name of Item	AI Query Interface
Description of Purpose	Allows a student to enter a natural-language college-related question.
Source of Input	Student

Destination of Output	HANSRAJ QUERY BRIDGE backend and AI/RAG processing pipeline.
Valid Range / Accuracy	Text shall be within the configured maximum query length. Empty or invalid submissions shall not be accepted.
Units of Measure	Characters for input length.
Timing	Query shall be submitted when the student selects “Send”.
Relationship to Other Inputs/Outputs	The submitted query is processed by the RAG system and may result in an AI answer, clarification request, or human-forwarding option.
Screen Format / Organization	Chat-style interface containing conversation history, query input box, Send button, and response area.
Window Format / Organization	Scrollable conversation window with the newest interaction displayed at the bottom.
Data Format	UTF-8 text.
Command Format	Natural-language query submitted using “Send/Enter”.
End Messages	Reliable answer → Grounded verified response displayed. Ambiguous query → Clarification requested by the system. Insufficient information → Raise Query option displayed.

C. AI Response Interface

Attribute	Specification
Name of Item	AI Response Interface
Description of Purpose	Displays the response generated using retrieved and verified Knowledge Base information.
Source of Input	AI/RAG backend.
Destination of Output	Student's query interface.
Valid Range / Accuracy	Response shall be grounded in sufficiently relevant and active Knowledge Base information. Unsupported answers shall not be presented as verified institutional information.
Units of Measure	Not applicable.
Timing	Response shall be displayed after completion of the retrieval and response-generation process.
Relationship to Other Inputs/Outputs	Directly associated with the student's submitted query and retrieved Knowledge Base information.
Screen Format / Organization	Response displayed as a separate message in the conversation interface.
Window Format / Organization	Responses shall appear chronologically with the corresponding user queries.
Data Format	UTF-8 text; structured formatting such as headings, lists, and links may be supported where appropriate.
Command Format	No direct command required; generated automatically.

End Messages	Answer displayed, or the interface provides an appropriate clarification message, or the interface provides an appropriate query forwarding message.
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D. Raise Query Interface

Attribute	Specification
Name of Item	Raise Query Interface
Description of Purpose	Allows a student to submit a query to the Admin when the AI cannot provide a reliable answer or when human assistance is requested.
Source of Input	Student
Destination of Output	Query Management subsystem and Admin dashboard.
Valid Range / Accuracy	Query and description shall satisfy configured length and validation rules; mandatory fields shall not be empty.
Units of Measure	Characters for text fields.
Timing	Submitted when the student selects Raise Query and confirms submission.
Relationship to Other Inputs/Outputs	Creates a query record that can subsequently be monitored, resolved by Admin, or forwarded to a TIC.
Screen Format / Organization	Form containing query/title, detailed description, submission controls, and validation messages.
Window Format / Organization	Dedicated form or dialog launched from the AI query interface.
Data Format	UTF-8 text; system-generated query ID and timestamps.
Command Format	Submit Query / Cancel.
End Messages	Successful submission → Confirmation containing query reference/status. Validation failure → Field-specific error message.

E. Query Tracking Interface

Attribute	Specification
Name of Item	Query Tracking Interface
Description of Purpose	Displays the current status, relevant details, and final resolution of a student's query.
Source of Input	Query Management subsystem and authorized Admin/TIC actions.
Destination of Output	Student dashboard page.
Valid Range / Accuracy	Status shall correspond to the current workflow state stored by the system.
Units of Measure	Date/time displayed using the configured institutional time-zone and format.
Timing	Status shall reflect the latest successfully stored workflow update.
Relationship to Other Inputs/Outputs	Changes when an Admin or TIC reviews, forwards, requests documents, or resolves a query.
Screen Format / Organization	Query ID, submitted query, current status, timeline/history, assigned authority where appropriate, and resolution information.

Window Format / Organization	Dedicated query-detail page or expandable dashboard view.
Data Format	Text, status values, timestamps, and system-generated identifiers.
Command Format	View Details / Refresh where applicable.
End Messages	Current status or final resolution displayed; unavailable information → Result in an appropriate error message.

F. Document Submission Interface

Attribute	Specification
Name of Item	Student Document Submission Interface
Description of Purpose	Allows a student to submit documents specifically requested by an authorized TIC.
Source of Input	Student's device and Student.
Destination of Output	Secure document-storage service and associated query record.
Valid Range / Accuracy	File type, file size, number of files, and other restrictions shall follow configured limits.
Units of Measure	File size in MB/GB; number of files.
Timing	Upload occurs when the student selects and submits the requested documents.
Relationship to Other Inputs/Outputs	Requires an active document request associated with the student's query. Successful submission updates the document status.
Screen Format / Organization	Requested-document list, file-selection control, upload button, submission status, and instructions.
Window Format / Organization	Dedicated document section within the query page or dashboard.
Data Format	Supported document/file formats as configured by the institution.
Command Format	Select File → Upload → Submit.
End Messages	Successful upload → Confirmation and status update; Invalid upload → Reason for rejection and corrective instruction.

G. Meeting Interface

Attribute	Specification
Name of Item	Meeting Interface
Description of Purpose	Allows students to view, accept, decline, or review meetings related to their queries.
Source of Input	Student and Timetable subsystem.
Destination of Output	Meeting Management subsystem and Student dashboard.
Valid Range / Accuracy	Only available and valid slots shall be presented for booking/acceptance.
Units of Measure	Date, time, and meeting details.
Timing	Meeting information shall be displayed when available and updated following scheduling actions.

Relationship to Other Inputs/Outputs	Meeting status is associated with the relevant student query and TIC.
Screen Format / Organization	Slot view containing date, time, mode/location, TIC, and meeting status.
Window Format / Organization	Meeting section on dashboard or query-details page.
Data Format	Date/time values, status values, and text details.
Command Format	Accept / Decline / View Meeting / Cancel where permitted.
End Messages	Meeting action confirmed and status updated; Unavailable slot → Appropriate conflict message.

H. Notification Interface

Attribute	Specification
Name of Item	Notification Interface
Description of Purpose	Provides users with system-generated notifications about important query, document, and meeting events.
Source of Input	Notification subsystem triggered by system workflows.
Destination of Output	Student, Admin, or TIC dashboard and configured email service.
Valid Range / Accuracy	Notifications shall be delivered only to the intended authorized recipient and shall correspond to an actual system event.
Units of Measure	Date/time for notification timestamp.
Timing	Notifications shall be generated following successful completion of the associated event, subject to configured delivery mechanisms.
Relationship to Other Inputs/Outputs	Triggered by actions such as query submission, forwarding, resolution, document requests/reviews, and meeting updates.
Screen Format / Organization	Notification icon/list containing event summary, timestamp, read/unread state, and link to the relevant record where permitted.
Window Format / Organization	Notification panel/drop-down.
Data Formats	Notification ID, message text, event type, timestamp, read/unread status, and related record reference.
Command Formats	View Notification, Mark as Read, Open Related Query/Meeting/Document.
End Messages	Notification displayed successfully; If no notifications exist, an empty-state message shall be shown.

I. Admin Query Management Interface

Attribute	Specification
Name of Item	Admin Query Management Interface
Description of Purpose	Allows Admins to inspect, filter, resolve, and forward student queries.
Source of Input	Query Management subsystem; Admin actions.

Destination of Output	Query database, TIC workflow, notification subsystem, and Student dashboard.
Valid Range / Accuracy	Only queries accessible to the authenticated Admin shall be displayed.
Units of Measure	Dates/times, query counts, and response durations where displayed.
Timing	Dashboard data shall reflect the latest stored query state.
Relationship to Other Inputs/Outputs	Admin actions directly change query status and may generate notifications or assignments.
Screen Format / Organization	Dashboard with query list, filters, search, status indicators, query details, and action controls.
Window Format / Organization	Main dashboard with query-detail panel for individual queries.
Data Format	Structured text, identifiers, status values, dates/times.
Command Format	View / Resolve / Forward / Request Clarification or other authorized actions.
End Messages	Action confirmation or validation/error message.

J. Admin Query Forwarding Interface

Attribute	Specification
Name of Item	Admin Query Forwarding Interface
Description of Purpose	Allows an authorized Admin to forward a query to the appropriate TIC when departmental intervention is required.
Source of Input	Admin and Query Management subsystem.
Destination of Output	Assigned TIC, Query Management subsystem, and Notification subsystem.
Valid Range / Accuracy	Only active and authorized TICs relevant to the query's department shall be selectable.
Units of Measure	Not applicable; timestamps recorded for forwarding actions.
Timing	Forwarding shall occur after the Admin confirms the selected TIC and submits the action.
Relationship to Other Inputs/Outputs	Changes the query status and assignment and triggers an appropriate notification to the TIC and, where applicable, the student.
Screen Format / Organization	Query details with department information, TIC selection, forwarding reason/notes, and confirmation controls.
Window Format / Organization	Forwarding section within the Admin query-detail page.
Data Formats	Query ID, TIC identifier, department, forwarding reason, timestamp, and status.
Command Formats	Select TIC, Forward Query, Cancel.
End Messages	Successful forwarding → Confirmation with assigned TIC and updated status. Invalid assignment → Error message.

K. Knowledge Base Management Interface

Attribute	Specification
Name of Item	Knowledge Base Management Interface

Description of Purpose	Allows authorized Admins to add, modify, deactivate, archive, and manage versions of verified institutional information.
Source of Input	Authorized Admin.
Destination of Output	Knowledge Base and vector-retrieval subsystem.
Valid Range / Accuracy	Information shall be institutionally verified and satisfy required metadata and content validation.
Units of Measure	Dates for validity periods; version numbers; text length where applicable.
Timing	Changes shall become available to the retrieval system after successful processing/indexing.
Relationship to Other Inputs/Outputs	Updates affect the information available to the AI/RAG query-resolution process.
Screen Format / Organization	Searchable list of entries with source, validity, version, status, and edit controls.
Window Format / Organization	Management dashboard with entry form/editor and version-history view.
Data Format	Structured text, metadata, dates, identifiers, and version information.
Command Format	Add / Update / Deactivate / Archive / View History.
End Messages	Successful update confirmation; validation error if required information is missing or invalid.

L. TIC Resolution Interface

Attribute	Specification
Name of Item	TIC Resolution Interface
Description of Purpose	Allows an authorized TIC to provide and submit a verified resolution for an assigned departmental query.
Source of Input	TIC.
Destination of Output	Query Management subsystem, Student Dashboard, Notification subsystem, and potentially the Knowledge Base workflow.
Valid Range / Accuracy	Resolution text shall satisfy configured input-length and validation requirements and shall contain sufficient information to address the query.
Units of Measure	Characters for resolution text; date/time for resolution timestamp.
Timing	Resolution shall be submitted when the TIC has completed the review of the query.
Relationship to Other Inputs/Outputs	Successful resolution changes the query status to Resolved and may trigger student notification and controlled Knowledge Base updating.
Screen Format / Organization	Query details, resolution text area, optional supporting information, Need Document option where applicable, and submission controls.
Window Format / Organization	Resolution form within the assigned-query detail page.

Data Formats	UTF-8 text, status values, timestamps, and optional references/attachments where supported.
Command Formats	Submit Resolution, Save Draft, Request Document, or other authorized actions.
End Messages	Resolution submitted successfully → Query status updated and notification initiated. Validation failure → Field-specific error message.

M. TIC Document Request and Review Interface

Attribute	Specification
Name of Item	Document Request and Review Interface
Description of Purpose	Allows a TIC to request required documents from a student and subsequently review documents submitted in response.
Source of Input	TIC; submitted documents originate from the Student interface.
Destination of Output	Document Management subsystem, Query Management subsystem, Student Dashboard, and Notification subsystem.
Valid Range / Accuracy	Document type, instructions, file requirements, and review status shall conform to configured institutional rules.
Units of Measure	File size in MB/GB; number of documents; timestamps for submission/review.
Timing	Request is created during query resolution; review occurs after the student submits the requested documents.
Relationship to Other Inputs/Outputs	A document request changes the query status to Documents Required . Student submission changes it to Documents Submitted . Review may result in acceptance, rejection, or a request for correction.
Screen Format / Organization	Query details, requested-document list, instructions, submitted files, document status, review controls, and reason/comment fields.
Window Format / Organization	Document section within the query-detail page, with separate request and review areas.
Data Formats	Document metadata, file formats supported by the system, status values, text comments, and timestamps.
Command Formats	Request Document, View Document, Accept, Reject, Request Correction, Submit Review.
End Messages	Request created → student notified. Review submitted → document status updated and student notified. Invalid/rejected file → reason displayed.

N. Timetable and Meeting Management Interface :

Attribute	Specification
Name of Item	Timetable and Meeting Management Interface
Description of Purpose	Allows TICs to view relevant timetable information and manage meeting slots with students.
Source of Input	TIC, Timetable/Calendar subsystem, and existing meeting records.

Destination of Output	TIC Dashboard, Meeting Management subsystem, Student Dashboard, and Notification subsystem.
Valid Range / Accuracy	Only valid and available meeting slots shall be offered. Existing bookings and timetable conflicts shall be considered before confirmation.
Units of Measure	Date, time, and meeting duration.
Timing	Timetable information shall be retrieved when requested; meeting status shall update after each successful scheduling action.
Relationship to Other Inputs/Outputs	Meeting information is associated with the relevant student/query and may be triggered after query review or document verification.
Screen Format / Organization	Timetable view displaying days, periods, occupied/available slots, selected student/query, meeting details, and action controls.
Window Format / Organization	Dedicated meeting-management page or section within the assigned-query interface.
Data Formats	Date/time values, meeting status, duration, mode/location, student information, and query ID.
Command Formats	View Timetable, Select Slot, Offer Meeting, Schedule, Reschedule, Cancel.
End Messages	Meeting successfully scheduled/offered → Confirmation and notification. Slot conflict → Booking rejected with an appropriate message.

3.1.2 Hardware Interfaces

HANSRAJ QUERY BRIDGE does not require any specialized hardware interfaces.

The system shall interact with hardware only through the standard operating-system and browser interfaces available on supported devices.

- **User Input Devices:** Keyboard, mouse, and touchscreen shall be used for entering queries, completing forms, navigating dashboards, uploading documents, and performing other system actions,
- **Display Interface:** The web application shall render its user interface through the display of the user's device and support responsive layouts.
- **Network Interface:** Client and server devices shall use their standard network interfaces to establish communication with HANSRAJ QUERY BRIDGE.

3.1.3 Software Interfaces

This section specifies the detailed interfaces between HANSRAJ QUERY BRIDGE and the software components required for its operation.

A. Frontend–Backend Interface

The frontend shall communicate with the FastAPI backend through RESTful APIs over HTTPS.

Attributes	Specification
Input	User requests, authentication, session information, query text, form data, status actions, meeting and document operations.
Output	Query results, AI responses, status information, validation messages, notifications, and operation confirmations.

Data Formats	JSON for structured data
Interface Behaviour	The backend shall validate requests, enforce authorization, process the requested operation, and return an appropriate HTTP response.

B. Authentication Interface

The authentication interface shall communicate with the configured college authentication/SSO service, where available.

Attributes	Specification
Input	Authentication request and required identity-provider credentials.
Output	Authentication result and authenticated user identity information.
Data Formats	Protocol and provider-dependent authentication messages and tokens.
Interface Behaviour	Successful authentication shall establish an authenticated session; invalid or expired authentication shall be rejected.

C. AI/RAG Interface

The backend shall communicate with the Gemini LLM and embedding services for AI-powered query processing.

Attributes	Specification
Input	Student query and relevant retrieved Knowledge Base context.
Output	Embeddings, retrieval-related results, and generated response.
Data Formats	UTF-8 text and model-specific API structures.
Interface Behaviour	Only sufficiently relevant verified Knowledge Base information shall be used for grounded responses. Queries that cannot be reliably answered shall be directed to the escalation workflow.

D. Database Interface

HANSRAJ QUERY BRIDGE shall communicate with PostgreSQL and pgvector for persistent and vector data operations.

Attributes	Specification
Input	Structured records, query parameters, embeddings, updates, and transaction requests.
Output	Stored records, retrieved records, similarity-search results, and operation status.
Data Formats	Relational table and vector representations.
Interface Behaviour	Database access shall occur through the backend and shall be restricted to authorized application services.

E. Document Storage Interface

The backend shall communicate with the configured secure file storage service for student documents.

Attributes	Specification
Input	Validated document files and associated metadata.
Output	Storage confirmation, document reference, or retrieval result.

Input	Supported file formats and metadata.
Output	Documents shall be accessible only to authorized users and shall be associated with the corresponding student and query.

F. Timetable Interface

Where an external timetable service is available, HANSRAJ QUERY BRIDGE shall communicate with it to obtain scheduling information.

Attributes	Specification
Input	Timetable queries and meeting-slot requests.
Output	Timetable information, availability, and scheduling status.
Data Formats	Provider-specific API data, including dates, times, and event information.
Interface Behaviour	The system shall verify slot availability and existing bookings before confirming a meeting.

G. Notification Interface

The notification subsystem shall communicate with the configured email service through RESEND.

Attributes	Specification
Input	Recipient information, notification type, and message content.
Output	Delivery status or service response.
Data Formats	Email message format or provider-specific API format.
Interface Behaviour	Notification events shall be recorded by HANSRAJ QUERY BRIDGE even if external email delivery fails.

3.1.4 Communication Interfaces

This section specifies the communication mechanisms used by HANSRAJ QUERY BRIDGE for exchanging data between clients, backend services, databases, and external services.

The system shall follow commonly supported communication standards and protocols, including:

Communication	Protocol / Format
Browser ↔ Backend	HTTPS / REST / JSON
Backend ↔ Database	Secure PostgreSQL connection
Backend ↔ AI Services	HTTPS / Gemini API
Backend ↔ File Storage	HTTPS / AWS S3 Bucket API
Backend ↔ Email Service	HTTPS / RESEND API
Backend ↔ Timetable Service	HTTPS / Provider API

A. Client – Server Communication

- The system shall use **HTTPS** for communication between the web browser and the HANS Connect backend.
- Requests and responses shall primarily use **RESTful HTTP** methods such as GET, POST, PUT/PATCH, and DELETE.

- Structured data shall be exchanged in **JSON** format.

B. Backed – Database Communication

- The backend shall communicate with PostgreSQL through a secure database connection.
- Database communication shall be restricted to authorized backend services.

C. File Transfer Communication

- Student documents shall be transferred through secure HTTPS-based mechanisms.
- File uploads shall be validated for permitted file type and size before storage.
- Sensitive documents shall not be transmitted through ordinary unsecured communication channels.

D. External Service Communication

HANSRAJ QUERY BRIDGE may communicate with configured external services such as:

- College authentication services
- LLM and embedding services
- Secure document storage
- Email notification services
- Timetable services

Such communication shall use the secure API/protocol specified by the respective service provider.

3.2 System Features

3.2.1 Authentication, Authorization and Role Management

A. Description

HANSRAJ QUERY BRIDGE shall provide secure user authentication and role-based authorization for **Students, TICs, and Admins**. The system shall verify the user's identity through the configured college authentication/SSO mechanism and assign the appropriate role and permissions. The system shall ensure that authenticated users can access only the functions and information permitted for their role.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	User login request	System redirects to authentication interface and displays the login page	Display Login screen
2	User credentials / SSO authentication	Authentication service verifies identity	Authentication result
3	Verified user identity	System retrieves user profile and assigned role	User identity + Assigned role
4	Role information	Authorization rules are applied	Display permitted dashboard and features
5	Authorized request	System processes requested operation	Requested resource/action
6	Logout request	System terminates the active session	Logged-out state

Sequence:

Login → Identity Verification → Role Identification → Authorization → Role-Specific Access → Logout

C. Validity Checks on Input

- Credentials/SSO authentication information shall be validated by the authentication provider.

- Required authentication fields shall not be empty.
- Expired or invalid authentication tokens shall be rejected.
- The user account shall be active.
- The assigned role shall be a valid system role: Student, TIC, or Admin.
- Requested resources/actions shall be checked against the user's permissions.

D. Input-to-Output Conversion

The functional mapping is:

Authentication Result = Verify(User Credentials/Identity)

If authentication is successful:

Authenticated Session = User Identity + Assigned Role + Session Validity

Access = Allow(User Role, Requested Resource) if Permission(User Role, Resource) = TRUE

Otherwise:

Access = Deny;

E. Sequence of Operation

1. User initiates login.
2. System authenticates the user through the authentication mechanism.
3. System verifies that the account is active.
4. System retrieves the user's role and institutional profile.
5. System creates an authenticated session.
6. System applies role-based access permissions.
7. User is directed to the appropriate dashboard.
8. Each subsequent protected operation is authorized before execution.
9. On logout or session expiration, the authenticated session is terminated.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Invalid credentials	Reject login and display an appropriate error message.
Authentication service unavailable	Inform the user that authentication is temporarily unavailable.
Expired session	Require the user to authenticate again.
Inactive user account	Deny access and display an appropriate message.
Invalid role	Deny access and display an appropriate message.
Unauthorized resource access	Reject the request with an authorization error.
Repeated failed login attempts	Apply configured security controls such as temporary blocking.
Logout failure	Invalidate the session server-side and prevent further authorized requests.

3.2.2 RAG-Based AI-Powered Query Resolution

A. Description

HANSRAJ QUERY BRIDGE shall use RAG to understand student queries, retrieve relevant information from the verified institutional Knowledge Base, and generate grounded responses. The system shall distinguish between different query intents and shall escalate queries to Admin when a reliable answer cannot be determined.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	Student's natural-language query	Validate and preprocess the query	Processed query
2	Processed query	Generate query embedding/semantic representation	Query vector
3	Query vector	Perform semantic similarity search against verified Knowledge Base	Relevant knowledge entries
4	Retrieved entries	Evaluate relevance and confidence	Relevant context + confidence score
5	Sufficient relevant context	Provide context to the LLM for grounded generation	AI-generated response
6	Insufficient/low-confidence context	Suppress unsupported response	Routing option
7	Generated response	Present answer with relevant institutional information	Student-facing answer

Sequence:

Student Query → Validation → Query Understanding → Embedding → Semantic Retrieval → Relevance Evaluation → Grounded Response Generation → Response OR Escalation

C. Input-to-Output Conversion

For a query embedding q and Knowledge Base entry embedding d , cosine similarity shall be used:

$$\text{Similarity}(q, d) = \frac{q \cdot d}{\|q\| \|d\|}$$

A confidence measure shall be represented as:

$$\text{Confidence} = f(\text{Similarity}, \text{Relevance}, \text{Source Validity})$$

where:

- Similarity = semantic similarity between the query and retrieved entries.
- Relevance = degree to which the retrieved information addresses the detected intent.
- Source Validity = whether the information comes from an active and verified Knowledge Base entry.

The system shall retrieve Knowledge Base entries whose similarity exceeds the retrieval threshold.

Output:

If Confidence $\geq T$: Response = Grounded Answer , where **T** is the configured confidence threshold.
Otherwise : Routing Option

D. Validity Checks on Input

- Query input shall not be empty.
- Input shall be sanitized to prevent malicious or malformed content.
- Retrieved information shall be evaluated for semantic relevance before being supplied to the response-generation component.
- Queries containing insufficient information may be answered only when the intended meaning can be determined reliably; otherwise, clarification or escalation shall be provided.

E. Sequence of Operation

1. Student submits a natural-language query.
2. System validates and preprocesses the input.
3. System determines the semantic representation/intent of the query.
4. System generates an embedding for the query.
5. System searches the verified Knowledge Base using semantic similarity.
6. System retrieves the most relevant valid entries.
7. System evaluates relevance and confidence.
8. If adequate information is available, the retrieved context is supplied to the LLM.
9. The LLM generates a response using the retrieved institutional information.
10. System presents the grounded response to the student.
11. If adequate information is unavailable, the system shall avoid unsupported generation and display the **Raise Query to Admin** option.
12. The resulting interaction and required metadata may be recorded for monitoring and future improvement.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Empty or invalid query	Prompt the student to enter a valid query.
No relevant Knowledge Base information	Offer the Raise Query to Admin option.
Low-confidence or ambiguous result	Do not provide a definitive answer; offer escalation.
Conflicting/outdated information	Use the current verified entry or escalate for review.
AI/RAG service unavailable	Inform the student and provide manual query escalation.

3.2.3 Knowledge Base Management**A. Description**

HANSRAJ QUERY BRIDGE shall provide an authorized Knowledge Base management facility that shall contain current information such as procedures, requirements, deadlines, contacts, departmental information, and frequently asked questions.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	Knowledge entry details	Validate the submitted information	Validated entry
2	New or updated information	Create or update the Knowledge Base entry	Updated entry
3	Verification action	Mark information as verified	Verified Knowledge Base entry
4	Deactivation/update request	Change the status or version of the entry	Updated entry status
5	Search/retrieval request	Retrieve relevant active entries	Knowledge Base results

Sequence:

Entry Submission → Validation → Verification → Storage/Update → Indexing → RAG Retrieval

C. Validity Checks on Input

- Required Knowledge Base fields shall not be empty.
- The information source shall be recorded.
- Each entry shall have a defined verification status.
- Each entry shall have a validity/current-status indicator.
- Duplicate or conflicting entries shall be identified before publication.
- Outdated information shall not remain active as authoritative information.
- Only authorized users shall be permitted to create, modify, verify, or deactivate entries.

D. Sequence of Operation

1. Admin shall enter or update institutional information.
2. System shall validate the submitted information.
3. System shall record the source and relevant metadata.
4. Authorized Admin shall verify the information.
5. System shall store the verified entry in the Knowledge Base.
6. System shall make the valid entry available for semantic retrieval.
7. When information changes, the existing entry shall be updated, replaced, or deactivated.

E. Abnormal Situations and Response

Abnormal Situation	System Response
Incomplete Knowledge Base entry	Reject submission and identify the missing information.
Unauthorized modification attempt	Deny the operation and record the event.
Outdated information	Deactivate or replace the entry with the current verified information.
Knowledge Base storage failure	Reject the update and preserve the existing valid data.

3.2.4 Intelligent Query Escalation and Routing**A. Description**

HANSRAJ QUERY BRIDGE shall identify queries that cannot be reliably resolved by the RAG system and shall escalate them for human resolution. The system shall route escalated queries to **Admin** and shall allow Admin to resolve the query or forward it to the appropriate **TIC** based on the query's department or subject.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	Low-confidence query	Identify need for escalation	Escalation request
2	Student query + description	Create query record	New query
3	New query	Assign to Admin	Admin notification and dashboard entry

4	Admin routing decision	Identify appropriate TIC or resolve directly	Resolved/forwarded query
5	Forwarded query	Assign query to selected TIC	TIC notification and dashboard entry
6	Resolution	Update query status and solution	Resolution available to student

Sequence:

Query → Confidence Check → Escalation → Admin → Resolve / Forward to TIC → Resolution → Student Notification

C. Input-to-Output Conversion

The escalation decision shall be based on the configured confidence threshold:

$$Escalate = \begin{cases} False, & Confidence \geq T \\ True, & Confidence < T \end{cases}$$

For routing:

$$Assigned\ TIC = Routing(Query, Department, Subject)$$

If no suitable TIC can be identified, the query shall remain with Admin for resolution.

D. Validity Checks on Input

- Query text shall not be empty.
- Escalated queries shall contain sufficient information for human review.
- The student's identity shall be associated with the query.
- The selected TIC shall be an active and authorized TIC.
- Query status shall be validated before routing or resolution.

E. Sequence of Operation

1. System shall evaluate the AI/RAG response confidence.
2. If the query cannot be reliably resolved, the system shall provide the **Raise Query to Admin** option.
3. Student shall submit the query with the required description.
4. System shall create and assign the query to Admin.
5. Admin shall review the query.
6. Admin shall either resolve the query or forward it to the appropriate TIC.
7. TIC shall review and resolve the assigned query.
8. System shall update the query status and notify the student.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Insufficient query details	Request additional information from the student.
No suitable TIC found	Keep the query assigned to Admin.
Invalid/inactive TIC	Prevent assignment and require another routing decision.

Routing operation fails	Preserve the query with its current owner/status and report the failure.
-------------------------	--

3.2.5 Role Specific Dashboards and Management Interfaces

A. Description

HANSRAJ QUERY BRIDGE shall provide separate dashboards and management interfaces for **Students, Admins, and TICs**. Each dashboard shall display information and actions relevant to the user's role and shall restrict access to unauthorized functions and data.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	Authenticated user	Identify user role	Role information
2	Role information	Load corresponding dashboard	Role-specific dashboard
3	User action/request	Validate authorization	Permitted operation
4	Operation result	Update relevant records	Updated dashboard/status
5	Navigation/action	Load selected management interface	Requested interface

Sequence:

Login → Role Identification → Dashboard Loading → Authorized Action → Data Update → Dashboard Refresh

C. Input-to-Output Conversion

Dashboard access shall be determined by the authenticated user's role:

$$\text{Dashboard} = f(\text{User Role})$$

Access to a specific operation shall be:

$$\text{Access} = \begin{cases} \text{Allow, } \text{Permission}(\text{Role}, \text{Action}) = \text{True} \\ \text{Deny, } \text{Permission}(\text{Role}, \text{Action}) = \text{False} \end{cases}$$

D. Validity Checks on Input

- User authentication shall be valid before dashboard access.
- The user's role shall be verified.
- Requested actions shall be checked against role permissions.
- Dashboard data shall be retrieved only for authorized users.

E. Sequence of Operation

1. User shall authenticate successfully.
2. System shall identify the user's assigned role.
3. System shall load the corresponding dashboard.
4. System shall display role-specific information and available actions.
5. User shall select an authorized operation.
6. System shall validate and execute the operation.
7. System shall update the relevant records.
8. Dashboard information shall be refreshed to reflect the updated state.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Invalid/expired session	Require re-authentication.
Unauthorized dashboard access	Deny access and display an appropriate error message.
Data retrieval failure	Display an error message and preserve existing records.
Session expires during an operation	Prevent unauthorized completion and require re-authentication.

3.2.6 Document Management

A. Description

HANSRAJ QUERY BRIDGE shall provide secure document management for queries that require supporting documents. TICs shall be able to request and review specific documents, and students shall be able to upload the requested documents through the system.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	Document requirement	TIC marks Need Document and specifies required documents	Document request
2	Document request	System notifies the student	Student document section/request
3	Selected document	System validates file type and size	Validated file
4	Valid document	System securely stores the document	Upload confirmation
5	Stored document	System associates it with the relevant query/student	Document record
6	TIC review action	System displays submitted document to authorized TIC	Review status/action
7	Review result	System updates document status	Updated status and notification

Sequence:

Document Request → Student Upload → Validation → Secure Storage → TIC Review → Status Update

C. Input-to-Output Conversion

Document validation shall be based on configured file requirements:

$$Valid\ Document = FileType_{valid} \wedge FileSize \leq MaxSize$$

Document association shall be:

$$Document\ Record = Student + Query + File + Metadata + Status$$

D. Validity Checks on Input

- A document shall be uploaded only for an active document request.
- File type shall be among the supported formats.
- File size shall not exceed the maximum limit.
- The student shall be authorized to upload the document for the associated query.
- Only authorized TICs shall be able to access submitted documents.

E. Sequence of Operation

1. TIC shall indicate that a document is required while resolving a query.
2. System shall record the document requirement and notify the student.
3. Student shall upload the required document.
4. System shall validate the uploaded file.
5. System shall securely store the valid document and associate it with the query.
6. TIC shall review the submitted document.
7. TIC shall accept the document, request another document, or continue the resolution process.
8. System shall update the document status and notify the student where required.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Unsupported file type	Reject the upload and display the allowed formats.
File exceeds size limit	Reject the upload and display the permitted size.
Invalid or corrupted file	Reject the upload and request a valid document.
Unauthorized document access	Deny access and record the attempt.
Document storage failure	Report the failure and retain the upload as incomplete.

3.2.7 Meeting and Timetable Management**A. Description**

HANSRAJ QUERY BRIDGE shall provide timetable and meeting management functionality for **TICs and Students**. TICs shall be able to view their available time slots and schedule meetings with students. Students shall be able to view their scheduled meetings through their dashboard.

B. Input/Output Sequences

Step	Input	System Operation	Output
1	TIC timetable	System retrieves available slots	Available meeting slots
2	Student/query details + selected slot	System checks slot availability	Validated booking request
3	Valid booking request	System creates meeting record	Meeting confirmation
4	Meeting confirmation	System updates student and TIC dashboards	Scheduled meeting
5	Meeting status	System updates meeting record	Updated meeting status

Sequence:

TIC Availability → View Slots → Select Slot → Availability Check → Booking → Confirmation → Dashboard Update

C. Input-to-Output Conversion

A meeting slot shall be considered available when it is within the TIC's available schedule and has no existing booking:

$$\text{Available Slot} = \text{Valid Schedule} \wedge \text{Slot Not Booked}$$

A meeting shall be confirmed when:

$$\text{Booking} = \text{Valid Student} \wedge \text{Valid TIC} \wedge \text{Available Slot}$$

D. Validity Checks on Input

- The selected TIC shall be active and authorized.
- The meeting date and time shall be valid.
- The student shall be associated with the relevant query or interaction.
- Required meeting details shall not be empty.

E. Sequence of Operation

1. TIC shall provide available meeting slots.
2. System shall display the available slots to the authorized user.
3. Student/TIC shall select a suitable slot.
4. System shall verify slot availability.
5. System shall create the meeting record upon successful validation.
6. System shall display the meeting in the relevant dashboards.
7. System shall send the required meeting notification to the student and TIC.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Invalid meeting date/time	Reject the booking and request valid details.
Timetable unavailable	Inform the user and prevent slot booking until availability is retrieved.
Booking operation fails	Preserve the existing schedule and report the failure.
Meeting record unavailable	Display an error and prevent inconsistent booking confirmation.

3.2.8 Notifications through Email**A. Description**

HANSRAJ QUERY BRIDGE shall send email notifications to students, Admins, and TICs for important system events such as query submission, query resolution, query forwarding, document requests, meeting scheduling, and relevant status updates.

B. Input / Output Sequences

Step	Input	System Operation	Output
1	System event	Identify notification type and recipient	Notification request
2	Recipient + message details	Generate email content	Formatted email
3	Formatted email	Send through configured email service	Delivery request
4	Email service response	Record delivery status	Delivery status / log

Sequence:

System Event → Identify Recipient → Generate Email → Send → Record Delivery Status

C. Input-to-Output Conversion

The recipient shall be determined from the event and associated user record:

$$\text{Recipient} = f(\text{Event}, \text{User}, \text{Role})$$

The notification shall be generated as:

$$\text{Email} = \text{Recipient} + \text{Subject} + \text{Message} + \text{Event Details}$$

D. Validity Checks on Input

- Recipient email address shall be valid and associated with the user account.
- Required notification details shall not be empty.
- The notification event shall correspond to a valid system action.
- Emails shall contain only information authorized for the intended recipient.

E. Sequence of Operation

1. A relevant system event shall occur.
2. System shall identify the intended recipient.
3. System shall generate the appropriate email using the event details.
4. System shall send the email through the configured email service.
5. System shall record the delivery status.
6. The related event shall remain available through the application's notification interface.

F. Abnormal Situations and Response

Abnormal Situation	System Response
Invalid recipient email	Reject the notification and record the failure.
Email service unavailable	Record the failure and retry according to configured rules.
Email delivery failure	Record the delivery failure without affecting the underlying system operation.
Missing notification details	Prevent email generation and log the incomplete event.

3.3 Performance Requirements

This section specifies the performance requirements that HANSRAJ QUERY BRIDGE needs to satisfy including static and dynamic state performance measures placed on the software or on human interactions.

3.3.1 Response Time Requirements

HANSRAJ QUERY BRIDGE shall meet the following response-time requirements under normal operating conditions:

Operation	Maximum Response Time
Login and dashboard loading	At most 5 seconds in normal conditions
Standard page navigation request	At most 3 seconds in normal conditions
Knowledge Base search	At most 2 seconds in normal conditions
RAG-based query response	At most 8 seconds in normal conditions
Query submission/escalation	At most 3 seconds in normal conditions
Document upload confirmation	At most 5 seconds in normal conditions
Meeting slot availability retrieval	At most 3 seconds in normal conditions
Meeting booking confirmation	At most 3 seconds in normal conditions
Notification generation	At most 5 seconds in normal conditions

Additional requirements:

- At least **95%** of standard user requests shall meet their specified response-time limit.
- RAG response time shall be measured from query submission to display of the generated response.
- Response-time measurements shall exclude delays caused solely by the user's network connection or unavailable external services.
- Failed or timed-out requests shall return an appropriate system response within **10 seconds**.

3.3.2 Concurrent User Requirements

HANSRAJ QUERY BRIDGE shall support the following concurrent-user requirements under normal operating conditions:

Operation	Requirement Type	Measurable Requirement
Registered user accounts	Static	Support at least 50,000 registered users
Simultaneously logged-in users	Dynamic	Support at least 500 concurrent users
Simultaneous AI queries	Dynamic	Support at least 100 concurrent AI queries
Simultaneous query submissions	Dynamic	Support at least 100 concurrent submissions
Simultaneous document uploads	Dynamic	Support at least 50 concurrent uploads
Simultaneous dashboard access	Dynamic	Support at least 300 concurrent dashboard sessions
Simultaneous meeting bookings	Dynamic	Support at least 50 concurrent booking requests

Additional requirements:

- The system shall maintain correct query, document, and meeting records under concurrent access.
- Concurrent operations shall not result in duplicate bookings or inconsistent query status.
- The system shall maintain the response-time requirements specified in **Section 3.3.1** for at least 95% of requests under the defined concurrent-user load.
- The above values shall be treated as minimum MVP capacity targets and shall be validated through performance testing.

3.3.3 Throughput Requirements

HANSRAJ QUERY BRIDGE shall support the following minimum throughput requirements under normal operating conditions:

Operation	Requirement Type	Measurable Requirement
API requests	Dynamic	At least 50 requests/second
RAG query processing	Dynamic	At least 10 AI queries/minute
Query submissions	Dynamic	At least 20 submissions/minute
Knowledge Base retrievals	Dynamic	At least 30 retrievals/second
Document uploads	Dynamic	At least 10 uploads/minute
Meeting booking requests	Dynamic	At least 20 bookings/minute
Email notification processing	Dynamic	At least 50 notifications/minute
Query records	Static	Support at least 100,000 stored query records

Additional requirements:

- Throughput shall be measured under the concurrent-user load specified in **Section 3.3.2**.
- The system shall process requests without data loss or duplication.
- Throughput shall be maintained while meeting the response-time requirements specified in **Section 3.3.1**.
- Throughput limits shall be configurable according to the capacity of deployed infrastructure and external services.

3.3.4 Storage Capacity Requirements

HANSRAJ QUERY BRIDGE shall provide sufficient storage for institutional records, Knowledge Base data, embeddings, documents, and system logs.

Storage Component	Requirement Type	Measurable Requirement
User and role records	Static	Support at least 10,000 user records
Query records	Static	Support at least 100,000 query records
Knowledge Base entries	Static	Support at least 100,000 verified entries

Vector/embedding records	Static	Support at least 100,000 embeddings
Uploaded documents	Static	Support at least 500 GB of document storage
Individual document size	Static	Maximum 15 MB per file
Meeting and timetable records	Static	Support at least 100,000 records
Notification records	Static	Support at least 500,000 records
Audit/system logs	Static	Support at least 4 GB per month
Database storage expansion	Dynamic	Storage shall be expandable without redesigning the application.

Additional requirements:

- The system shall maintain sufficient free storage capacity to prevent failure of critical operations.
- Document storage and database storage shall be managed separately where required.
- Storage limits shall be configurable according to deployment requirements.
- The system shall reject files exceeding the configured maximum size.
- Storage usage shall be monitored to identify capacity-related risks.

3.3.5 Service Response Requirements

HANSRAJ QUERY BRIDGE shall maintain the following service-level requirements:

Service Requirement	Requirement Type	Measurable Requirement
System availability	Dynamic	At least 99% monthly uptime except maintenance
Service monitoring	Dynamic	System health shall be monitored at least every 60 seconds
Critical service failure detection	Dynamic	Failure shall be detected within 10 minutes
Critical service recovery	Dynamic	Service shall be restored within 30 minutes of a detected critical failure
Database backup	Dynamic	Backups shall be performed at least once every 24 hours
Data recovery point	Static/Dynamic	Maximum acceptable data loss shall be 24 hours
Planned maintenance notification	Static	Users shall be notified at least 12 hours before planned downtime

Additional requirements:

- The system shall continue serving unaffected functions when a non-critical external service fails.
- Service failures shall be logged for monitoring and troubleshooting.
- Critical failures shall trigger an appropriate administrative alert.

3.4 Logical Database Requirements

This section specifies the logical database requirements that HANSRAJ QUERY BRIDGE needs to satisfy for the storage of any information that is to be placed in a database.

3.4.1 Data Entities and Their Relationships

HANSRAJ QUERY BRIDGE shall maintain logically structured data entities required for user management, query resolution, Knowledge Base management, document handling, meetings, notifications, and system monitoring.

A. Primary Data Entities

Entity	Purpose
User	Stores student, Admin, and TIC account information.
Role	Defines permissions associated with each user type.
Department	Stores institutional department information.
Query	Stores student queries and their resolution status.
Query Assignment	Records Admin/TIC assignment and routing information.
Knowledge Base Entry	Stores verified institutional information used by the RAG system.
Document	Stores metadata and references to uploaded student documents.
Meeting	Stores student-TIC meeting information and booking status.
Timetable/Availability (External Service)	TIC availability and meeting slots may be retrieved from the external timetable service
Notification	Stores notification events and delivery status.
AUDIT_LOG	Records important system and administrative actions.

B. Entity Relationships

Relationship	Cardinality
User → Role	Many-to-One
User → Department	Many-to-One
Student → Query	One-to-Many
Department → Query	One-to-Many
Query → Knowledge Base Entry	One-to-One
Query → Query Assignment	One-to-Many
Admin/TIC → Query Assignment	One-to-Many
Department → TIC	One-to-Many
Query → Document Request	One-to-Many
Document Request → Document	One-to-Many

Query → Meeting	One-to-Many
TIC → Timetable/Availability	One-to-Many
User → Notification	One-to-Many
Query → Notification	One-to-Many
User → Audit Record	One-to-Many
Document & Meetings → Notification	One-to-Many

C. Relationship Requirements

- Each User shall have one valid primary role.
- Each Student query shall be associated with the student who submitted it.
- Each escalated query shall have an associated assignment record.
- Each TIC assignment shall reference an active TIC and the relevant department.
- Each uploaded document shall be associated with the relevant student and query.
- Each meeting shall reference the participating student and TIC.
- Each Knowledge Base entry shall have associated verification and validity information.
- Each notification shall reference its intended recipient and triggering event.
- Important system actions shall be associated with an audit record.

3.4.2 Data Integrity Constraints

HANSRAJ QUERY BRIDGE shall maintain accurate, consistent, and reliable data throughout its database operations.

A. Entity Integrity

- Each database entity shall have a unique primary key.
- Primary key values shall not be null or duplicated.
- System-generated identifiers (Surrogate Keys) shall remain unique throughout their lifetime.

B. Referential Integrity

- Foreign keys shall reference valid records in the corresponding parent entity.
- Queries shall reference valid users and departments where applicable.
- Documents shall reference valid queries and users.
- Meetings shall reference valid students and TICs.
- Notifications shall reference valid recipients.

C. Data Consistency

- User role and permission information shall remain consistent across related records
- Query status shall remain consistent with its assignment and resolution state.
- A Knowledge Base entry marked as active shall contain valid verification information.
- Document and query statuses shall reflect their current workflow state.

D. Transactional Integrity

- Related database changes shall be executed as atomic transactions where required

- Failed transactions shall be rolled back without leaving partial records.
- Concurrent operations shall not produce duplicate or conflicting records.

Critical operations such as query assignment, meeting booking, and Knowledge Base updates shall maintain consistent database state.

3.4.3 Knowledge Base Requirements

The Knowledge Base shall store **verified and current institutional information** used by the RAG system to answer student queries. It shall support semantic retrieval, controlled updates, validity management, and traceability.

A. Knowledge Base Content

- The system shall store institutional information such as procedures, requirements, deadlines, contacts, departments, FAQs, notices, and guidelines.
- Each Knowledge Base entry shall contain:
 - Entry ID
 - Title/Question
 - Answer/Content
 - Category
 - Department, where applicable
 - Source
 - Verification status
 - Validity period, where applicable if required
 - Status Information
 - Created at/Updated at timestamp
 - Verified By
- The Knowledge Base shall support at least **50,000 verified entries**.
- Each entry shall be uniquely identifiable.

B. Verification, Validity and Versioning

- Only verified and active entries shall be available for automatic RAG-based responses.
- Status being Unverified, expired, or deactivated entries shall not be used for answering queries.
- Each entry shall have a verification status indicating whether it is approved for retrieval.
- The system shall maintain updated at timestamps for time-sensitive information.
- Significant updates shall create a new version while preserving previous versions for traceability.
- Conflicting or inconsistent information shall be flagged for administrative review.
- Knowledge Base modifications shall be restricted to authorized users.

C. Semantic Retrieval Requirements

- Knowledge Base entries shall support embedding-based semantic retrieval.
- Retrieval shall be based on semantic relevance and shall not depend on exact keyword matching alone.
- The system shall retrieve the most relevant verified entries for a student query.
- The retrieval process shall distinguish between similar but different query intents.
- Retrieved information shall provide sufficient context for grounded response generation.
- If no sufficiently relevant verified information is available, the system shall not generate an unsupported institutional answer and shall follow the human escalation workflow.

D. Knowledge Base Access Control and Maintenance

- Students shall have read-only access to Knowledge Base information through the query-resolution interface.
- Students shall not directly modify Knowledge Base entries.
- Admin users shall be authorized to add, update, verify, deactivate, and review Knowledge Base entries.
- Authorized TIC users shall update Knowledge Base information only when the required permission is assigned.
- The system shall record Knowledge Base changes with the user, timestamp, action, and affected entry.
- New or corrected institutional information shall be incorporated into the Knowledge Base without requiring model retraining.
- The system shall allow outdated information to be replaced or deactivated while retaining historical records.

3.4.4 Documents Data Storage Requirements

The system shall securely store **documents submitted by students** as part of the query-resolution workflow. Document storage shall maintain confidentiality, integrity, accessibility, and association with the relevant student and query.

A. Document Storage Content

- The system shall store uploaded documents in secure AWS S3 bucket rather than directly in the application database.
- The database shall store document metadata and the reference to the stored file.
- Each document shall have a unique Document ID.
- Each document shall be associated with:
 - Student
 - Query ID
 - Document Request ID, where applicable
 - Document type
 - File name
 - File format
 - File size
 - Upload timestamp
 - Upload status
 - Review status
 - Storage Key
 - Review Comments
 - Reviewed At information

B. Storage Capacity and File Limits

- The system shall support **at least 500 GB** of uploaded document storage.
- The maximum individual document size shall **be 25 MB per file**.
- The system shall support commonly required document formats : PDF, JPG, JPEG, and PNG.
- Unsupported file types shall be rejected.
- Storage capacity shall be expandable without requiring changes to the application logic.

C. Document Association and Integrity

- Every uploaded document shall be associated with the correct student and query
- The system shall prevent documents from being associated with non-existent or unauthorized records.
- Uploaded files shall be validated for file type and size before storage.
- The system shall maintain the document's metadata consistently with its stored file.

D. Documents Access Control, Security and Retention

- Students shall access only documents they are authorized to view.
- TICs shall access documents only for queries assigned to them.
- Admin users shall access documents according to their assigned permissions.
- Unauthorized document access shall be denied and logged.
- Direct public access to uploaded documents shall be prohibited.
- The system shall retain document metadata and review status for traceability.
- Documents shall be retained according to the institution's retention policy.

3.4.5 Vector and Embedding Data Requirements

The system shall store and manage vector embeddings of Knowledge Base content to support **semantic search and RAG-based query retrieval**. Embedding data shall remain consistent with the corresponding Knowledge Base content and shall support efficient retrieval.

A. Embedding Generation

- Each active Knowledge Base entry shall have an associated embedding.
- The system shall generate an embedding when a new verified entry is added.
- The system shall regenerate the embedding when the associated content is significantly modified.
- Each embedding shall store:
 - KB Entry ID
 - KB Version
 - Embedding Vector Information
- **Embedding representation** : $E = f(\text{KB Content})$
where E is the generated vector and f is the configured embedding model.

B. Vector Dimensions and Storage

- The embedding dimension shall be fixed for a given embedding model.
- All vectors within the same vector index shall have the same dimension.
- The system shall support at least 100,000 embedding records.
- Vector data shall be stored using pgvector or a compatible vector database mechanism.
- Changing the embedding model shall require regeneration of affected embeddings.

C. Semantic Similarity and Retrieval

- The student's query shall be converted into a query embedding: $Eq = f(\text{Student Query})$
- Cosine similarity shall be supported:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

- Similarity Score shall range from **-1 to 1**, with higher values representing greater similarity.
- The system shall retrieve the **Top-K relevant entries**, where K shall be configurable, with an initial value of **5**.
- The threshold shall be adjustable based on evaluation results to reduce incorrect or weak matches.

D. Vector Data Integrity

- Only **verified and active** KB entries shall participate in vector retrieval.
- Every active vector shall reference exactly one valid KB entry version.
- Orphaned or invalid vectors shall be excluded from retrieval.

3.5 Design Constraints

The design of HANSRAJ QUERY BRIDGE shall comply with technical, institutional, hardware, software, and infrastructure constraints that affect system implementation and operation.

- **Availability of Student Data:** The college shall provide access to a reliable and sufficiently updated student database containing required information such as student identity, department, year, and institutional credentials.
The absence or inaccuracy of this data shall restrict authentication and student-specific services.
- **Institutional Authentication:** Availability of the college's existing authentication/SSO mechanism shall be required for institutional login.
If such a service is unavailable or access is not provided, an alternative authentication mechanism shall be required.
- **Knowledge Base Availability:** A sufficiently complete and verified institutional Knowledge Base shall be required for reliable AI-based query resolution.
Limited or outdated institutional information shall reduce the system's ability to answer queries automatically.
- **TIC and Department Information:** Current information regarding departments, TIC-Department assignments, responsibilities, and contact details shall be required for accurate query routing.
- **Timetable and Meeting Information:** Up-to-date TIC timetables and available meeting slots shall be required for the meeting-scheduling functionality.
Missing or outdated schedules shall restrict accurate slot allocation.
- **Institutional Policies:** The system shall operate within the college's existing policies for student records, document handling, query resolution, communication, privacy and data retention.
Changes in institutional procedures shall require corresponding system configuration or Knowledge Base updates.
- **Document Handling:** The system shall be constrained by the institution's permitted document types, maximum file sizes, retention requirements, and access policies.
- **External Service Availability:** AI services, email services, authentication services, and other required external services shall be subject to their availability, usage limits, and service interruptions.
- **Network and Infrastructure:** System availability and response time shall depend on the availability and capacity of the college or user network and server infrastructure.
- **Data Accuracy:** The quality of system outputs shall depend on the accuracy and currency of institutional data supplied to the system.
Incorrect source data can result in incorrect routing, notifications, or AI responses.

3.5.1 Standards Compliance

HANSRAJ QUERY BRIDGE shall follow applicable software, web accessibility, security, data-protection, and institutional standards to ensure consistent, secure, and auditable system operation.

- The system shall follow the **IEEE Std 830-1998 guidelines for Software Requirements Specifications** in documenting and maintaining its software requirements.
- The system's web interface shall follow applicable **WCAG 2.1 Level AA accessibility guidelines** to support users with different abilities.

- Personal data shall be collected, processed, stored, and deleted in accordance with the **Digital Personal Data Protection Act, 2023 (DPDP Act)** and applicable institutional privacy policies.
- Authentication and access control shall follow established secure identity and access-management practices, ensuring that users can access only authorized resources.
- System records shall follow consistent data naming, date/time, status, and identification conventions to maintain data consistency.
- Important system activities shall maintain an audit trail containing the responsible user, action, affected record, and timestamp.
- System-generated reports and records shall use consistent formats and predefined status values to support institutional administration and reporting.
- Any additional standards or regulations mandated by the college or applicable authorities shall be incorporated into the system requirements before deployment.

3.6 Software System Attributes

This section specifies the quality attributes and operational characteristics that HANSRAJ QUERY BRIDGE shall maintain in addition to its functional requirements.

These attributes define how effectively the system performs under different conditions, including increased users, changing workloads, failures, and future expansion.

3.6.1 Scalability

HANSRAJ QUERY BRIDGE shall be designed to accommodate growth in users, queries, Knowledge Base content, documents, and AI/RAG workload without requiring a complete redesign of the system.

- The system shall support **at least 10,000 registered users** and **500 concurrent users** while maintaining the performance requirements defined in Section 3.3.
- The system shall support **at least 100 concurrent AI queries** and **100,000 query records**.
- The Knowledge Base shall support at least **50,000 verified entries** and **100,000 embedding records**.
- Document storage shall support **at least 500 GB** and shall be independently expandable as required.
- The application, database, vector storage, and document storage shall support vertical or horizontal scaling to accommodate increased workload.
- The system shall support future increases in capacity without requiring major changes to the core application architecture.

3.6.2 Reliability

HANSRAJ QUERY BRIDGE shall provide consistent and dependable operation by minimizing failures, preventing data loss, and maintaining correct system state during normal and abnormal conditions.

- The system shall ensure transaction integrity for critical operations such as query submission, query assignment, document submission, and meeting booking, preventing duplicate or inconsistent records.
- The system shall detect critical system failures within **10 minutes** and initiate recovery procedures within **30 minutes**, where technically feasible.
- Temporary failures of external services such as AI, email, or storage shall be handled without losing existing user or query data.
- Failed operations shall provide appropriate error handling and shall be logged for **diagnosis and recovery**.

3.6.3 Availability

HANSRAJ QUERY BRIDGE shall remain accessible and operational for users during the defined service period. Availability shall be maintained through monitoring, checkpoints, backup, recovery, and controlled restart mechanisms.

- The system shall maintain **99% monthly availability**, excluding scheduled maintenance.
- The system shall perform automated database and configuration backups **at least once every 24 hours** and maintain recovery checkpoints for critical system data.
- The system shall continuously monitor critical services and shall detect major **failures within 10 minutes**.
- After a critical failure, the system shall restart affected services and restore normal operation within **30 minutes**, where technically feasible.
- The system shall prevent data loss during restart by recovering from the latest valid database state or backup checkpoint.
- Planned maintenance shall be scheduled during low-usage periods and communicated to users **at least 12 hours in advance**.
- If a non-critical external service such as email or AI becomes unavailable, core functions such as query submission and status tracking shall remain operational wherever possible.

3.6.4 Usability

HANSRAJ QUERY BRIDGE shall provide an intuitive interface that minimizes the effort required for users to learn, operate, provide input, and understand system output.

- The system shall provide **role-specific dashboards** for Students, TICs, and Admins with only the functions relevant to each role.
- A new student shall be able to perform basic tasks such as asking a query, raising a query, checking its status, and viewing a resolution **without formal training**.
- The AI query interface **shall accept natural-language queries**, reducing the need for users to learn specific commands or formats.
- Forms shall provide clear labels, required-field indicators, input validation, and meaningful error messages to reduce incorrect input.
- AI responses, query statuses, notifications, and resolutions shall be presented in **clear and understandable language** with relevant supporting information.
- The interface shall provide consistent navigation, terminology, buttons, and status indicators across the system.
- The system shall provide immediate feedback for important actions such as query submission, document upload, meeting booking, and resolution updates.
- The interface shall be responsive and usable.

3.6.5 Maintainability

HANSRAJ QUERY BRIDGE shall be designed so that errors can be identified, diagnosed, corrected, tested, and deployed with minimal disruption to ongoing services.

- The system shall maintain **centralized application, database, API, and AI/RAG logs** containing sufficient information to identify the source of errors.
- **Errors shall be recorded** with relevant details such as timestamp, affected component, operation, error type, and status, without exposing sensitive user data.

- The system shall **provide meaningful error messages** to users while keeping technical error details restricted to authorized administrators/developers.
- Fixes shall be tested in a separate environment before deployment to the production system.
- The system shall **maintain version-controlled source code** and configuration to support rollback to a previous stable version when a deployment introduces an error.

3.6.6 Privacy

HANSRAJ QUERY BRIDGE shall protect the privacy of students, TICs, Admins, and other individuals whose personal information is processed by the system. The system shall collect, process, store, and disclose personal data only for legitimate institutional purposes and in accordance with applicable privacy laws and college policies.

- The system shall collect only the **minimum personal information** required for authentication, query resolution, document processing, communication, and meeting management.
- Personal data and sensitive documents shall be protected during transmission and storage using appropriate security mechanisms such as **HTTPS/TLS and encrypted storage** where supported.
- Personal information shall not be used for purposes unrelated to the operation of HANSRAJ QUERY BRIDGE without appropriate authorization.
- The system shall maintain appropriate retention and deletion policies for personal data and uploaded documents, based on institutional requirements and applicable law.
- The system shall comply with applicable Indian data-protection requirements, including the **Digital Personal Data Protection Act, 2023 (DPDP Act)**, as applicable to the institution and its processing activities.

3.6.7 Security

HANSRAJ QUERY BRIDGE shall protect system resources and institutional data from unauthorized access, modification, disclosure, and misuse through authentication, authorization, secure communication, and access-control mechanisms.

- The system shall enforce **RBAC** for Students, TICs, and Admins, ensuring that users can access only the functions and data permitted for their role.
- The system shall authenticate users through college **SSO/OAuth** where available; locally managed credentials, if required, shall be securely stored and never retained in plaintext.
- All client-server communication shall use **HTTPS/TLS**, and authentication sessions/tokens shall have controlled validity and secure expiration.
- The system shall prevent the AI/RAG component from exposing restricted information or bypassing established access-control rules.

3.7 Other Requirements

This section specifies additional requirements that support the continuity, recovery, traceability, deployment, and long-term operation of HANSRAJ QUERY BRIDGE and are not fully covered by the functional requirements or software system attributes.

3.7.1 Accessibility Requirements

HANSRAJ QUERY BRIDGE shall be accessible and usable by users with different abilities and across commonly used devices and browsers.

- The system shall provide **keyboard-accessible navigation** and controls for all major functions.
- The interface shall use **clear labels, readable text, meaningful error messages, and consistent navigation**.
- Form fields, buttons, document-upload controls, and interactive elements shall provide appropriate labels and focus indicators.
- The system shall follow applicable **WCAG 2.1 Level AA** accessibility principles wherever reasonably applicable.

3.7.2 Traceability Requirements

HANSRAJ QUERY BRIDGE shall maintain sufficient audit records to trace important system activities, changes, and workflow decisions. Audit data shall support accountability, troubleshooting, and institutional review.

- The system shall record important actions such as login attempts, query creation, forwarding, resolution, document access, meeting booking, Knowledge Base changes, and permission changes.
- Audit records shall be protected from unauthorized modification or deletion.
- The system shall allow authorized Admin users to search and filter audit records by user, action, record, and date/time.
- Audit logs shall not unnecessarily store sensitive information such as passwords, authentication tokens, or complete document contents.
- Audit records shall be retained according to the institution's data-retention and compliance requirements.

3.7.3 AI and RAG Requirements

HANSRAJ QUERY BRIDGE shall apply AI and RAG techniques in a controlled, institution-specific manner. The AI component shall prioritize correct intent identification and appropriate handling of ambiguity before generating or escalating a response.

A. Query Understanding and Intent Matching

- The system shall convert each valid student query into a semantic representation for intent analysis rather than relying only on keyword matching.
- The system shall use a intent-confidence score:

$$C_{intent} = \max P(i | Q)$$

where Q is the student query and I is the set of supported institutional intents.

- Intent matching shall consider the query's meaning, context, entities, and required information, where available.
- For ambiguous queries, the system shall request clarification or follow the defined escalation process instead of producing a potentially incorrect institutional answer.

B. Information Retrieval

- The system shall retrieve the Top-K relevant Knowledge Base entries, with K = 5 as the initial configuration.
- Retrieval relevance shall be determined using the similarity score defined in Section 3.3.6:

$$S(q, d) = \frac{Eq \cdot Ed}{\|Eq\| \|Ed\|}$$

where E_q is the query embedding and E_d is the Knowledge Base entry embedding.

- The system shall prioritize the highest-relevance entries when constructing the context supplied to the LLM.

C. Confidence and Response Generation

- The system shall calculate an overall confidence score using relevant factors such as intent confidence, retrieval similarity, source validity, and consistency:

$$C_{overall} = w1C_{intent} + w2C_{retrieval} + w3C_{validity} + w4C_{consistency}$$

Where $w1 + w2 + w3 + w4 = 1$

- The generated response shall be restricted to information supported by the retrieved context and shall not introduce unsupported institutional facts. If no retrieved entry satisfies the relevance requirement, the system shall not use unrelated Knowledge Base content merely to produce an answer.
- The system shall clearly indicate when information is unavailable or insufficient instead of presenting an uncertain answer as factual.
- AI-generated responses shall be stored with sufficient metadata, including the query, retrieved source IDs, confidence score, and response status, for evaluation and traceability.

D. Human-in-the-Loop and Continuous Improvement

- Queries with low confidence, unresolved intent, conflicting information, or no suitable retrieved answer shall be routed to Admin for human resolution.
- Admin shall resolve the query directly or forward it to the appropriate TIC when department-specific expertise is required.
- New Knowledge Base content shall be embedded and indexed for future retrieval without requiring retraining of the underlying LLM.
- Human corrections shall take precedence over previous AI-generated responses when updating verified institutional information.

E. AI Safety and Robustness

- The AI shall not override RBAC, document permissions, or institutional access controls based on instructions contained in a student query.
- AI-generated responses shall not expose passwords, authentication tokens, private student information, internal system instructions, or unauthorized documents.
- AI service failures, timeouts, or malformed responses shall be detected and handled without corrupting query records.

APPENDIX

A. System Architecture Diagram

The following system architecture diagram was presented at the SIH Internal Hackathon Round on 25th August 2026, and shall be the reference architecture model for the HANSRAJ QUERY BRIDGE project.

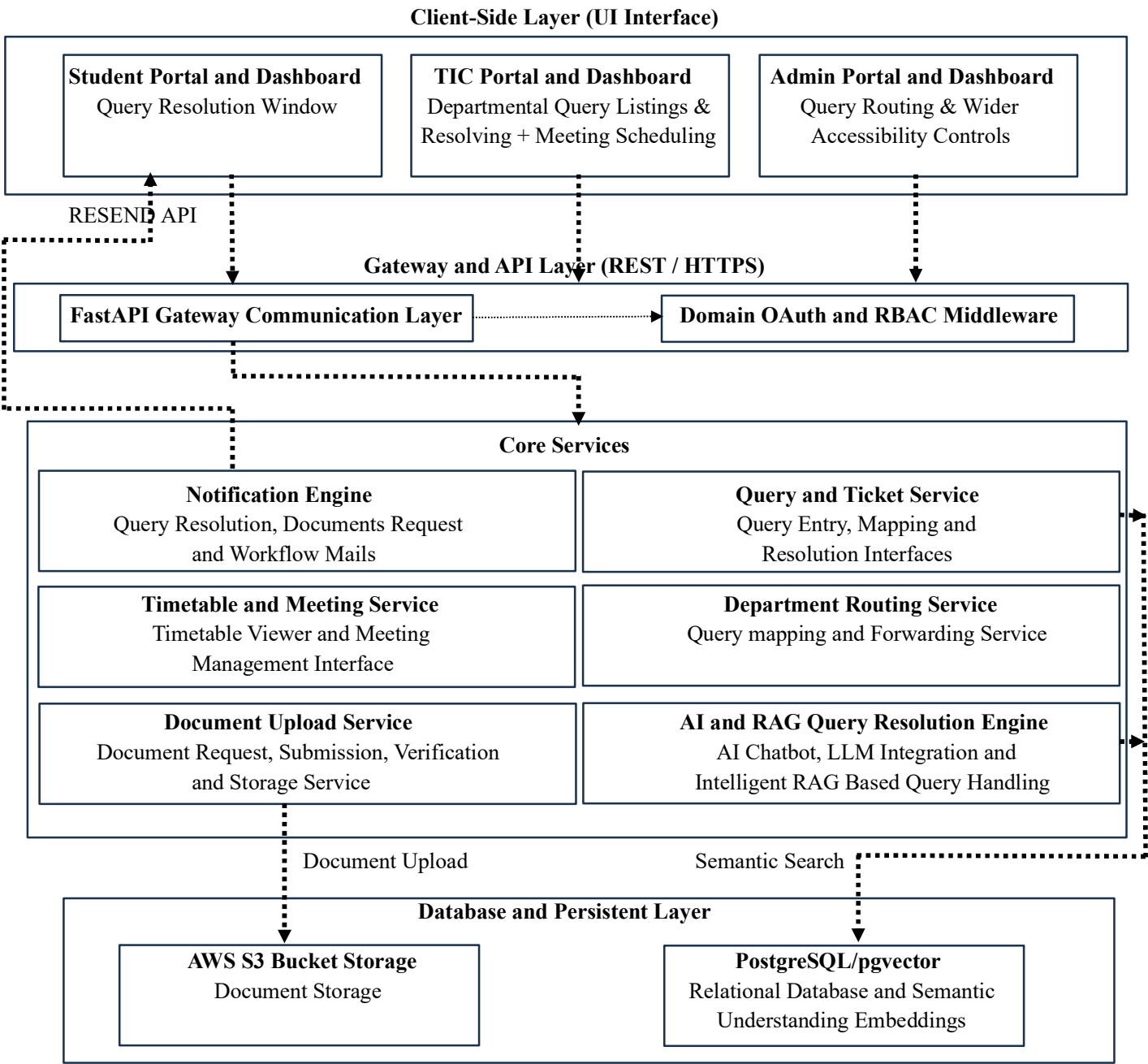


Figure A.1 : HANSRAJ QUERY BRIDGE : System Architecture Diagram

B. AI/RAG Flow

The following AI/RAG Flow Chart was presented at the SIH Internal Hackathon Round on 25th August 2026, and shall be the reference flow for the HANSRAJ QUERY BRIDGE project.

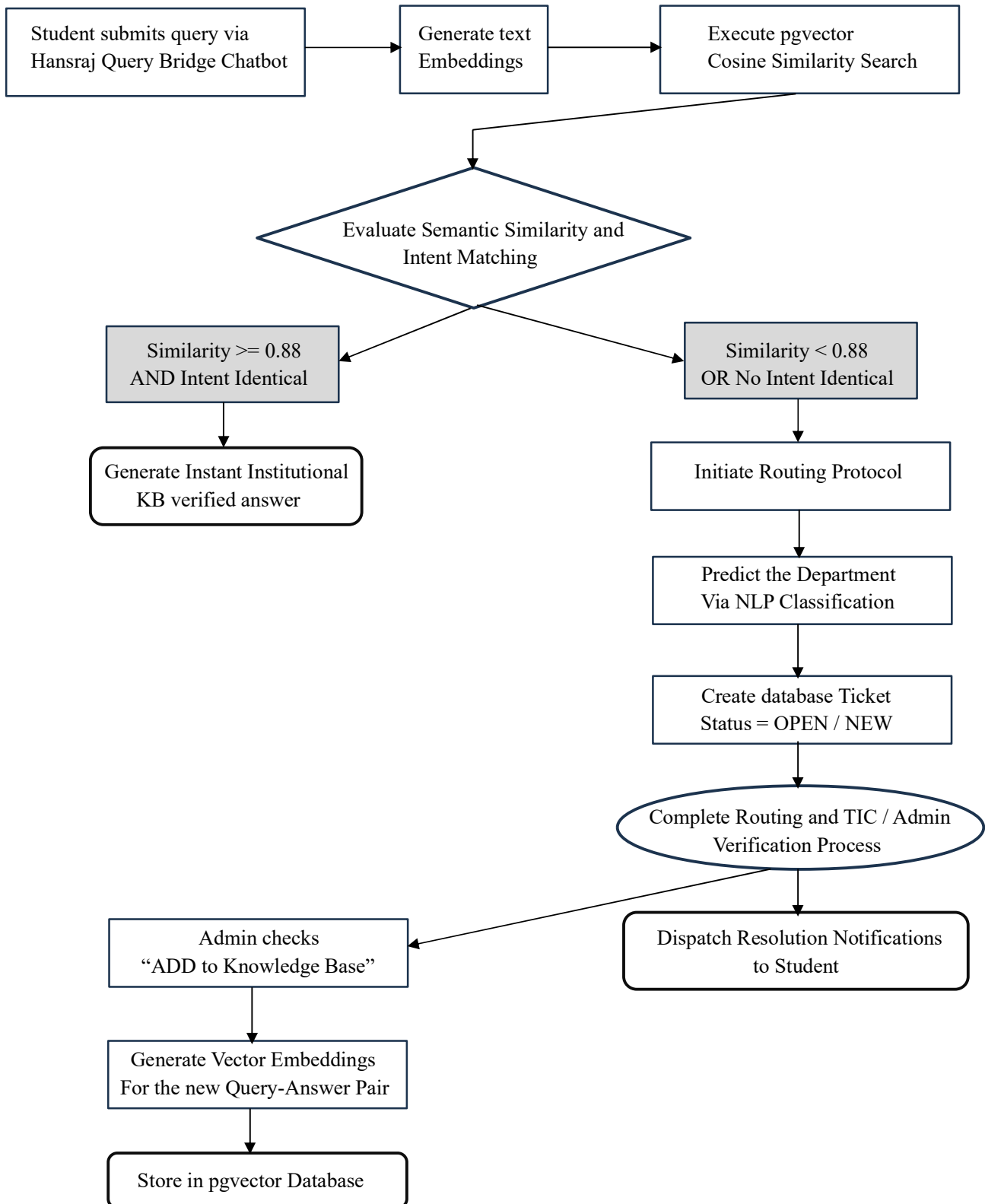


Figure A.2 : HANSRAJ QUERY BRIDGE : AI/RAG Overview Flow Diagram